

Math 40 Notes

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1 Matrices and Matrix Operations

Definition 1.1. An $m \times n$ matrix is a rectangular array of real numbers arranged in m horizontal rows and n vertical columns.

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}$$

Definition 1.2. An $m \times n$ matrix where $m, n \in \mathbb{R}^+$ and $m = n$, wherein it has the same number of rows and columns is called a **square matrix**. A $1 \times n$ matrix is called a **row vector**. And an $n \times 1$ matrix is called a **column vector**.

Definition 1.3. We will define the set of all $m \times n$ matrices with $\mathbb{R}$ entries as $M_{m \times n}(\mathbb{R})$. And we will denote the set of all square matrices with size n as $M_n(\mathbb{R})$.

Example 1.1. Let A be a square matrix with size 3×3 , and

$$A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

Find the value of the following entries: a_{12} , a_{31} , and a_{33} .

Recall that a_{ij} is the (i, j) entry of A where i represents the row and j is the column. For the example above use the subscript ij as a guide. a_{12} is the $(1, 2)$ entry in the matrix, thus, $a_{12} = 1$. The following entries are left to the reader.

Example 1.2. Let B be a matrix wherein $B = [b_{ij}]_{3 \times 2}$ where $b_{ij} = i + j$, draw the matrix and its entries.

$$B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \\ 4 & 5 \end{bmatrix}$$

The b_{ij} entries are left as an exercise to the reader.

Remark 1.1. Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal, denoted by $A = B$, if they have the same size and $a_{ij} = b_{ij}$ for all i, j .

However, by [Remark 1.1](#), this does not imply that two matrices with the same size are equal. Let us illustrate an example. Suppose that there exists two matrices $C, D \in M_n(\mathbb{R})$.

$$C = \begin{bmatrix} -4 & 6 \\ \textcircled{1} & \textcircled{7} \end{bmatrix}, \quad D = \begin{bmatrix} -4 & 6 \\ \textcircled{7} & \textcircled{1} \end{bmatrix}$$

Notice that they have different $(2, 1)$ and $(2, 2)$ entries. Thus, $[c_{ij}] \neq [d_{ij}]$.

Definition 1.4. Let $A = [a_{ij}]_{n \times n}$, wherein

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \in M_n(\mathbb{R}),$$

1. a_{ij} where $i = j$ are called the main diagonal entries
2. A is said to be:
 - a. **diagonal** if all entries $a_{ij} = 0$ where $i \neq j$,
 - b. **upper triangular** if all entries $a_{ij} = 0$ where $i > j$,
 - c. **lower triangular** if all entries $a_{ij} = 0$ where $i < j$.

Definition 1.5. An $n \times n$ matrix is called an **identity matrix**, denoted by I_n wherein all the diagonal entries are equal to 1, and the rest are 0.

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & \cdots & 1 \end{bmatrix} \in M_n(\mathbb{R})$$

An $m \times n$ **zero matrix**, denoted by $0_{m \times n}$ is a matrix wherein all its entries are 0.

Remark 1.2. Listed below are some observations on the matrices we have discussed:

1. $I_n = [\delta_{ij}]_{n \times n}$ where it is represented by a piecewise function¹ $\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$
2. Suppose you have a square matrix of size n , $A \in M_n(\mathbb{R})$ is diagonal with diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$, then we write

$$A = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$$

To illustrate examples of diagonal matrices:

$$P = \text{diag}(2, \pi, e, 0) = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & \pi & 0 & 0 \\ 0 & 0 & e & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$I_2 = \text{diag}(1, 1) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$0_{3 \times 3} = \text{diag}(0, 0, 0) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Do note that the notation for diagonal matrices **only applies to square matrices**. And in general, suppose there exists an identity matrix I_n it can be represented with the diagonal notation with $\text{diag}(\underbrace{1, 1, \dots, 1}_n)$ (this also applies to the zero matrix but replace 1 with 0).

¹Also known as the Kronecker delta function.

Definition 1.6. Let us consider two matrices with $m \times n$ size $A = [a_{ij}]$ and $B = [b_{ij}]$, $A, B \in M_{m \times n}(\mathbb{R})$ and a scalar $c \in \mathbb{R}$

1. The sum of A and B ,

$$A + B = [a_{ij} + b_{ij}]_{m \times n}$$

2. The scalar product of c and A ,

$$cA = [ca_{ij}]_{m \times n}$$

3. The transpose of A ,

$$A^T = [t_{ij}]_{n \times m}$$

where $t_{ij} = a_{ji}$.

Example 1.3. Let us illustrate the definitions we have shown above:

- 1.

$$\begin{aligned} \begin{bmatrix} 4 & -3 \\ 6 & 0 \\ 7 & -1 \end{bmatrix} + 3 \begin{bmatrix} 1 & 1 \\ -2 & 5 \\ 0 & -2 \end{bmatrix} &= \begin{bmatrix} 4 & -3 \\ 6 & 0 \\ 7 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 3 \\ -6 & 15 \\ 0 & -6 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 0 \\ 0 & 15 \\ 7 & -7 \end{bmatrix} \end{aligned}$$

- 2.

$$\begin{bmatrix} -1 & 0 \\ 4 & 1 \\ 2 & -3 \end{bmatrix}^T = \begin{bmatrix} -1 & 4 & 2 \\ 0 & 1 & -3 \end{bmatrix}$$

- 3.

$$\begin{bmatrix} -1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}^T = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Notice that in this example the transpose of a matrix is equal to itself. We call such matrices **symmetric**, we will further expand on this on [Chapter 14](#).

- 4.

$$\frac{1}{2} [4 \ 2 \ 0]^T - \begin{bmatrix} 1 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix}$$

Theorem 1.1 (Properties of Matrix Addition, Scalar Multiplication, and Transposition).

Let $A, B, C \in M_{m \times n}(\mathbb{R})$ and $c, d \in \mathbb{R}$. Then,

1. Matrix addition is commutative and associative,

$$A + B = B + A$$

$$A + (B + C) = (A + B) + C$$

2. There exists an identity matrix $0_{m \times n}$ such that when added to A it returns A ,

$$A + 0_{m \times n} = A$$

3. There exists an additive inverse $-A = -1(A)$, such that when added to A it returns $0_{m \times n}$,

$$A + (-A) = 0_{m \times n}$$

4. Scalars could commute,

$$(cd)A = c(dA) = d(cA)$$

5. Scalars could be distributed,

$$(c + d)A = cA + dA$$

$$c(A + B) = cA + cB$$

6.

$$(A + B)^T = A^T + B^T$$

7.

$$(cA)^T = cA^T$$

8.

$$(A^T)^T = A$$

Example 1.4. Let $A = \begin{bmatrix} -1 & 1 \\ 2 & -3 \\ 4 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 5 & 4 \\ 3 & -2 \end{bmatrix}$. Solve for X in the matrix equation $4X^T - 3A = B$.

$$\begin{aligned} 4X^T = B + 3A &\implies X^T = \frac{1}{4}(B + 3A) \\ &\implies (X^T)^T = \left[\frac{1}{4}(B + 3A) \right]^T \\ X = \left[\frac{1}{4}B + \frac{3}{4}A \right]^T &= \left[\begin{bmatrix} 0 & 1/4 \\ 5/4 & 1 \\ 3/4 & -1/2 \end{bmatrix} + \begin{bmatrix} -3/4 & 3/4 \\ 3/2 & -9/4 \\ 3 & -15/4 \end{bmatrix} \right]^T \\ &= \begin{bmatrix} -3/4 & 11/4 & 15/4 \\ 1 & -5/4 & -17/4 \end{bmatrix} \end{aligned}$$

Definition 1.7. Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$, we define a product of A and B , denoted by

$$A \times B = [c_{ij}]_{m \times p} \text{ where } c_{ij} = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix} \begin{bmatrix} b_{1j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Example 1.5. Find AB and BA ²

1.

$$A = \underbrace{\begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \end{bmatrix}}_{2 \times 3}, \quad B = \underbrace{\begin{bmatrix} -2 & 5 \\ 4 & -3 \\ 2 & 1 \end{bmatrix}}_{3 \times 2}$$

(a)

$$\begin{aligned} AB &= \begin{bmatrix} 1(-2) + 2(4) + (-1)(2) & 1(5) + 2(-3) + (-1)(1) \\ 3(-2) + 1(4) + (-1)(2) & 3(5) + 1(-3) + 4(1) \end{bmatrix} \\ &= \begin{bmatrix} 4 & -2 \\ 6 & 16 \end{bmatrix} \end{aligned}$$

²The underbraces under the matrices are simply visuals and should not be placed when solving.

(b)

$$\begin{aligned} BA &= \begin{bmatrix} -2 & 5 \\ 4 & -3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -2(1) + 5(3) & -2(2) + 5(1) & -2(-1) + 5(4) \\ 4(1) + (-3)(3) & 4(2) + (-3)(1) & 4(-1) + (-3)(4) \\ 2(1) + 1(3) & 2(2) + 1(1) & 2(-1) + 1(4) \end{bmatrix} \\ &= \begin{bmatrix} 13 & 1 & 22 \\ -5 & 5 & -16 \\ 5 & 5 & 2 \end{bmatrix} \end{aligned}$$

2.

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

(a)

$$\begin{aligned} AB &= \begin{bmatrix} 1(-1) + 2(1) & 1(2) + 2(-1) \\ 1(-1) + 1(1) & 1(2) + 1(-1) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

(b)

$$\begin{aligned} BA &= \begin{bmatrix} -1(1) + 2(1) & -1(2) + 2(1) \\ 1(1) + (-1)(1) & 1(2) + (-1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Theorem 1.2. Let A, B, C be matrices of suitable size and $c \in \mathbb{R}$,

1. If $A \in M_{m \times n}(\mathbb{R})$ then

(a) $I_m A = A = A I_n$

(b) $A 0_{n \times p} = 0_{m \times p}$ and $0_{q \times m} A = 0_{q \times n}$

2. $(AB)C = A(BC)$

3. $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$

4. $c(AB) = (cA)B = A(cB)$

5. $(AB)^\top = B^\top A^\top$

2 Solving Systems of Linear Equations

Consider a system of linear equations with n unknowns and m equations:

$$(*) = \begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = b_m \end{cases}$$

Theorem 2.1. For $(*)$ exactly one of the following holds:

- $(*)$ has exactly one solution (consistent and independent)
- $(*)$ has infinitely many solutions (consistent and dependent)
- $(*)$ has no solutions (inconsistent)

Example 2.1. We could rewrite $(*)$ into its matrix form, $Ax = b$, where A is the coefficient matrix and b is the vector of constants,

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_x = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_b$$

and we can denote the augmented matrix $[A \mid b]$ as

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_n \end{array} \right]$$

Notice that we have placed the coefficients in the left side and is separated by a vertical bar.³ On the right side we have the constants.

Example 2.2. Consider the following system of equations

$$\begin{cases} w + 3x + y + z = 3 \\ 2w + 2x + y + 2z = 8 \\ 3w + x + 2y - z = -1 \end{cases}$$

We can rewrite this in matrix form as:

$$\begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 2 & 1 & 2 \\ 3 & 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \\ -1 \end{bmatrix}$$

Shown below is its form in augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 1 & 2 & 8 \\ 3 & 1 & 2 & -1 & -1 \end{array} \right]$$

³This serves as a visual representation on what we're separating.

Recall

In your previous math classes (Pre-calculus), you have learned solving system of equations through elimination by addition. To solve a system of equations, the goal is to convert it to an easier form that is easy to solve.

Question. Which processes will yield equivalent system of equations?

Suppose we have the following system of equations:

$$\begin{cases} x - y = -2 \\ 2x + 6y = 12 \end{cases}$$

I. Switching the position of the equations:

$$\begin{cases} x - y = -2 \\ 2x + 6y = 12 \end{cases} \longleftrightarrow \begin{cases} 2x + 6y = 12 \\ x - y = -2 \end{cases}$$

II. Multiplying by a nonzero constant,

$$\begin{cases} x - y = -2 \\ 2x + 6y = 12 \end{cases} \longleftrightarrow \begin{cases} 2x + 6y = 12 \\ x - y = -2 \end{cases}$$

III. Add a constant multiple of one equation to another

$$\begin{aligned} \begin{cases} x - y = -2 \\ 2x + 6y = 12 \end{cases} &\longleftrightarrow \begin{cases} x - y = -2 \\ 8y = 16 \end{cases} \\ &\longleftrightarrow \begin{cases} x - y = -2 \\ y = 2 \end{cases} \end{aligned}$$

All of these operations will yield the equivalent system of equations!

Suppose we take the system of equations from above and convert it to an augmented matrix,

$$\left[\begin{array}{cc|c} 1 & 1 & -2 \\ 0 & 1 & 2 \end{array} \right]$$

Notice that it has a staircase form shown, as

$$\left[\begin{array}{cc|c} \boxed{1} & 1 & -2 \\ 0 & \boxed{1} & 2 \end{array} \right]$$

This form is also known as the **row echelon form**.

Definition 2.1. An $m \times n$ matrix is said to be in **row echelon form** if the following holds:

- All zero rows of A (if any) are at the bottom of A ;
- The leftmost nonzero entry of every nonzero row of A is 1;
- The leading entry of a higher row is strictly to the left of the leading entry of the next row.

Remark 2.1. $A \in M_{m \times n}(\mathbb{R})$ is in RREF (reduced row echelon form) if A is in REF (reduced echelon form) and every leading entry of A is the only nonzero entry of its column.

Remark 2.2. Each equation in a system corresponds to a row in its augmented matrix. The following processes will now yield equivalent systems:

- **Type I** - Switch two rows in augmented matrix;
- **Type II** - Multiply a row in augmented matrix by nonzero constant;
- **Type III** - Add a constant multiple of one row in the augmented matrix to another row.

2.1 Gauss-Jordan Reduction

To solve the system $Ax = b$:

- Form the augmented matrix $[A \mid b]$
- Use elementary row operations to bring augmented matrix to RREF
- Express the resulting matrix as a system of linear equations
- Use back-substitution

Remark 2.3. Gaussian elimination is the term when dealing with matrices in REF form.

Example 2.3. Use Gauss-Jordan Reduction to find the solution set of the following equations:

$$\begin{cases} 2x + 3y + z = 2 \\ 2x + 4y + 2z = 8 \\ x - y - z = 4 \end{cases}$$

Solution.

Form the augmented matrix and apply elementary row operations,

$$\begin{aligned} & \begin{bmatrix} 2 & 3 & 1 & | & 2 \\ 2 & 4 & 2 & | & 8 \\ 1 & -1 & -1 & | & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & -1 & -1 & | & 4 \\ 2 & 4 & 2 & | & 8 \\ 2 & 3 & 1 & | & 2 \end{bmatrix} \\ & \xrightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1}} \begin{bmatrix} 1 & -1 & -1 & | & 4 \\ 0 & 6 & 4 & | & 0 \\ 0 & 5 & 3 & | & -6 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_3} \begin{bmatrix} 1 & -1 & -1 & | & 4 \\ 0 & 1 & 1 & | & 6 \\ 0 & 5 & 3 & | & -6 \end{bmatrix} \\ & \xrightarrow{R_3 \rightarrow R_3 - 5R_2} \begin{bmatrix} 1 & -1 & -1 & | & 4 \\ 0 & 1 & 1 & | & 6 \\ 0 & 0 & -2 & | & -36 \end{bmatrix} \xrightarrow{R_3 \rightarrow -\frac{1}{2}R_3} \begin{bmatrix} 1 & -1 & -1 & | & 4 \\ 0 & 1 & 1 & | & 6 \\ 0 & 0 & 1 & | & 18 \end{bmatrix} \\ & \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{bmatrix} 1 & 0 & 0 & | & 10 \\ 0 & 1 & 1 & | & 6 \\ 0 & 0 & 1 & | & 18 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_3} \begin{bmatrix} 1 & 0 & 0 & | & 10 \\ 0 & 1 & 0 & | & -12 \\ 0 & 0 & 1 & | & 18 \end{bmatrix} \\ & \implies \begin{cases} x = 10 \\ y = -12 \\ z = 18 \end{cases} \end{aligned}$$

Therefore, the solution set is $\boxed{\{(10, -12, 18)\}}$.

Example 2.4. Use Gauss-Jordan Reduction to find the solution set of the following equations:

$$\begin{cases} w + x + 2y + 3z = 13 \\ w - 2x + y + z = 8 \\ 3w + x + y - z = 1 \end{cases}$$

Solution.

Form the augmented matrix and apply elementary row operations,

$$\begin{aligned} & \begin{bmatrix} 1 & 1 & 2 & 3 & | & 13 \\ 1 & -2 & 1 & 1 & | & 8 \\ 3 & 1 & 1 & -1 & | & 1 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \begin{bmatrix} 1 & 1 & 2 & 3 & | & 13 \\ 0 & -3 & -1 & -2 & | & -5 \\ 0 & -2 & -5 & -10 & | & -38 \end{bmatrix} \\ & \xrightarrow{R_2 \rightarrow R_2 - R_3} \begin{bmatrix} 1 & 1 & 2 & 3 & | & 13 \\ 0 & -1 & 4 & 8 & | & 33 \\ 0 & -2 & -5 & -10 & | & -38 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{bmatrix} 1 & 1 & 2 & 3 & | & 13 \\ 0 & -1 & 4 & 8 & | & 33 \\ 0 & 0 & -13 & -26 & | & -104 \end{bmatrix} \\ & \xrightarrow{\substack{R_2 \rightarrow -R_2 \\ R_3 \rightarrow -\frac{1}{13}R_3}} \begin{bmatrix} 1 & 1 & 2 & 3 & | & 13 \\ 0 & 1 & -4 & -8 & | & -33 \\ 0 & 0 & 1 & 2 & | & 8 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 0 & 6 & 11 & | & 46 \\ 0 & 1 & -4 & -8 & | & -33 \\ 0 & 0 & 1 & 2 & | & 8 \end{bmatrix} \\ & \xrightarrow{\substack{R_1 \rightarrow R_1 - 6R_3 \\ R_2 \rightarrow R_2 + 4R_3}} \begin{bmatrix} 1 & 0 & 0 & -1 & | & -2 \\ 0 & 1 & 0 & 0 & | & -1 \\ 0 & 0 & 1 & 2 & | & 8 \end{bmatrix} \Rightarrow \begin{cases} w - z = -2 \\ x = -1 \\ y + 2z = 8 \end{cases} \end{aligned}$$

Let $z = t$, then, $w = t - 2$, $y = 8 - 2t$. Therefore, the solution set is $\boxed{\{(t - 2, -1, -2t + 8, t) \mid t \in \mathbb{R}\}}$.

Remark 2.4.

1. The variable z in the system of equations above is called a **free variable**.
2. A system $Ax = b$ having at least one solution has k free variables, where k is the number of columns in the RREF of $[A \mid b]$ having no leading entries. The system is said to have a k -parameter solution.

For the rest of the examples, I will not show the whole process.⁴ It is up to the reader to verify that the solution is correct.

Example 2.5. Use Gaussian Elimination to find the solution set.

$$\begin{cases} x - y + 2z = 4 \\ x + z = 6 \\ 2x - 3y + 5z = 4 \\ 3x + 2y - z = 1 \end{cases}$$

Solution.

Form the augmented matrix and apply elementary row operations,

$$\begin{bmatrix} 1 & -1 & 2 & | & 4 \\ 1 & 0 & 1 & | & 6 \\ 2 & -3 & 5 & | & 4 \\ 3 & 2 & -1 & | & 1 \end{bmatrix} \xrightarrow{\text{REF}} \begin{bmatrix} 1 & -1 & 2 & | & 4 \\ 0 & 1 & -1 & | & 2 \\ 0 & 0 & 1 & | & 21/2 \\ 0 & 0 & 0 & | & 1 \end{bmatrix}$$

⁴Since this is TOO tedious to format.

$$\implies \begin{cases} x - y + 2z = 4 \\ y - z = 2 \\ z = 21/2 \\ 0 = 1 \implies \text{This is a contradiction!} \end{cases}$$

Since the solution is inconsistent, therefore, the solution set is $\emptyset$ or $\{\}$.

Remark 2.5. The system $Ax = b$ has no solution iff $[0 \ 0 \ \cdots \ 0 \ | \ 1]$ is a row in an REF of $[A \ | \ b]$.

Example 2.6. Solve for the solution of the following system of equations.

$$\begin{cases} 2x_1 + x_2 - 4x_3 - 5x_4 + 5x_5 = 5 \\ 3x_1 - x_2 - 6x_3 - 5x_4 = 5 \\ 5x_1 - 2x_2 - 10x_3 - 8x_4 - x_5 = 8 \end{cases}$$

Solution.

Form the augmented matrix and apply elementary row operations,

$$\left[\begin{array}{ccccc|c} 2 & 1 & -4 & -5 & 5 & 5 \\ 3 & -1 & -6 & -5 & 0 & 5 \\ 5 & -2 & -10 & -8 & -1 & 8 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccccc|c} 1 & -2 & -2 & 0 & -5 & 0 \\ 0 & 1 & 0 & -1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\implies \begin{cases} x_1 - 2x_2 - 2x_3 - 5x_5 = 0 \\ x_2 - x_4 + 3x_5 = 1 \end{cases}$$

Let $x_3 = r, x_4 = s, x_5 = t$, then, $x_2 = s - 3t + 1$ and $x_1 = 2r + 2s - t + 2$. Therefore, the solution set is $\boxed{\{(2r + 2s - t + 2, s - 3t + 1, r, s, t) \mid r, s, t \in \mathbb{R}\}}$.

3 Inverse of a Square Matrix

Recall that for all $a \in \mathbb{R}$, there exists a multiplicative inverse $a^{-1} \in \mathbb{R}$, such that

$$a \cdot a^{-1} = 1 = a^{-1} \cdot a$$

Now let us figure out if $0_{n \times n} + A \in M_n(\mathbb{R})$, can we find a $B \in M_n(\mathbb{R})$ such that $AB = I_n = BA$? The answer to that is **not always!**

As a counterexample, suppose that we have two matrices A and B , such that their product $AB = I_2$,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Now suppose this is true, then,

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This is a contradiction, because if we compare the $(2, 2)$ entries of both matrices, $0 \neq 1$. Hence no such B exists.

Definition 3.1. A square matrix $A \in M_n(\mathbb{R})$ is said to be **nonsingular/invertible** if there exists a matrix $B \in M_n(\mathbb{R})$ such that

$$AB = I_n = BA$$

In this case, B is said to be the multiplicative inverse of A . If A has no inverse, it is said to be **singular/invertible**.

Theorem 3.1. The inverse of a nonsingular matrix is unique.

Proof. Assume for the sake of contradiction that A has two inverse B and C , such that

$$AB = BA = I_n$$

$$AC = CA = I_n$$

Recall that B can be expressed as

$$\begin{aligned} B &= BI_n = B(AC) \\ &= (BA)C = I_n C = C \end{aligned}$$

Because $B = C$, which is a contradiction, therefore, the inverse is unique. ■

Notation: Let $A \in M_n(\mathbb{R})$,

1. If A is nonsingular, then its unique inverse is denoted by A^{-1}
2. If $A \neq 0_n$ then $A^0 = I_n$
3. For any $k \in \mathbb{Z}^+$, we represented the matrix being multiplied k times as

$$A^k = \underbrace{A \times A \times \cdots \times A}_k$$

4. The laws of exponents also apply,

$$A^k \cdot A^l = A^{k+l}$$

$$(A^k)^l = A^{kl}$$

However,

$$(AB)^2 \neq A^2B^2$$

Remark 3.1. (Properties of Nonsingular Matrices).

Let $A, B \in M_n(\mathbb{R})$ and nonsingular, $0 \in \mathbb{R}$, and $k \in \mathbb{Z}^+$. Then the following are nonsingular:

1. A with its inverse A^{-1}
2. AB with its inverse⁵ $(AB)^{-1} = B^{-1}A^{-1}$
3. cA with its inverse $\frac{1}{c} \cdot A^{-1}$
4. $A^\top$ with its inverse $(A^\top)^{-1} = (A^{-1})^\top$
5. A^k with its inverse $(A^k)^{-1} = (A^{-1})^k$

Theorem 3.2 (Cancellation Laws). Let B and C be matrices of suitable sizes and $A \in M_n(\mathbb{R})$ be nonsingular. Then,

$$AB = AC \implies B = C$$

$$BA = CA \implies B = C$$

Proof. Since A is nonsingular and its inverse A^{-1} exists, then suppose that $AB = AC$ is true, we can show that by multiplying both sides by its inverse,

$$A^{-1}(AB) = A^{-1}(AC)$$

$$I_n B = I_n C$$

$$B = C$$

Similarly we can show that $B = C$ when $BA = CA$,

$$(BA)A^{-1} = (CA)A^{-1}$$

$$BI_n = CI_n$$

$$B = C$$

■

Remark 3.2. If $A, B \in M_n(\mathbb{R})$ such that $AB = I_n$, then $BA = I_n$ and so A is nonsingular.

To further understand this remark, let us put it into an example. Suppose $A \in M_n(\mathbb{R})$ such that $A^2 - 2A + I_2 = 0$, figure out whether A is nonsingular.

Solution.

$$\begin{aligned} I_2 &= 2A - A^2 \\ &= (2I_2)(A) - AA \\ &= \underbrace{(2I_2 - A)}_B(A) \end{aligned}$$

Therefore, A is nonsingular with inverse $A^{-1} = 2I_2 - A$.

⁵This is also known as the [socks-shoes principle](#).

Fact 3.1. If A and B are matrices of size $n \times n$ and $b_j = \text{col } j \text{ of } B$, then

$$AB = A \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} Ab_1 & Ab_2 & \cdots & Ab_n \end{bmatrix}$$

Theorem 3.3. $A \in M_n(\mathbb{R})$ is nonsingular iff the system of equations, $Ax = b$ has a unique solution for any b , given by

$$x = A^{-1}b$$

Proof. Suppose that A is nonsingular and let $b \in M_{n \times 1}(\mathbb{R})$. Take $x = A^{-1}b$,

$$A(A^{-1}b) = I_n b = b$$

Therefore, $Ax = b$ has at least one solution. Let us use proof by contradiction to show that the system has a unique solution. Suppose then that x_1 and x_2 are solutions of $Ax = b$ such that

$$Ax_1 = b = Ax_2$$

By cancellation law, we can show that, $x_1 = x_2$. Which is a contradiction, hence, the system $Ax = b$ has only one solution. Suppose $Ax = b$ has exactly one solution for any $b \in M_{n \times 1}(\mathbb{R})$, if $e_j = \text{col } j$ of I_n , then $Ax = e_j$ has a unique solution x_j .

Let $B = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$,

$$\begin{aligned} AB &= A \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \\ &= \begin{bmatrix} Ax_1 & Ax_2 & \cdots & Ax_n \end{bmatrix} \\ &= \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix} = I_n \end{aligned}$$

Therefore, A is nonsingular with inverse B . ■

Remark 3.3. Let $A \in M_n(\mathbb{R})$ be nonsingular. Then column j of A^{-1} is the unique solution to $Ax = e_j$ where e_j is column j of I_n .

We will abbreviate the elementary row operations in some texts as **EROs**.

- $A \in M_n(\mathbb{R})$ is nonsingular if there exists an inverse $A^{-1} \in M_n(\mathbb{R})$ such that

$$A \cdot A^{-1} = I_n = A^{-1} \cdot A$$

- To find A^{-1}

$$[A \mid I_n] \xrightarrow{\text{EROs}} [C \mid D]$$

wherein $C = I_n$ and $D = A^{-1}$

- If C has a row of 0s, then A is singular

Remark 3.4. $A \in M_n(\mathbb{R})$ is nonsingular iff A is row equivalent to I_n , and A is said to be singular iff A is row equivalent to a matrix of zero row.

Definition 3.2. A matrix $E \in M_n(\mathbb{R})$ is called an **elementary matrix** if it can be obtained by applying a single elementary row operation on I_n .

Example 3.1. Let $A = \begin{bmatrix} 2 & -4 & 6 \\ 3 & -5 & 8 \\ 0 & 1 & -4 \end{bmatrix}$, use elementary row operations to find the inverse A^{-1} .

Solution.

Form the augmented matrix,

$$\left[\begin{array}{ccc|ccc} 2 & -4 & 6 & 1 & 0 & 0 \\ 3 & -5 & 8 & 0 & 1 & 0 \\ 0 & 1 & -4 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 5/3 & 1/3 \\ 0 & 1 & 0 & -2 & 4/3 & -1/3 \\ 0 & 0 & 1 & -1/2 & 1/3 & -1/3 \end{array} \right]$$

$$\therefore A^{-1} = \begin{bmatrix} -2 & 5/3 & 1/3 \\ -2 & 4/3 & -1/3 \\ -1/2 & 1/3 & -1/3 \end{bmatrix}$$

Example 3.2. Use the answer from [Example 3.1](#) to find the unique solution of the system of equations:

$$\begin{cases} 2x - 4y + 6z = 1 \\ 3x - 5y + 8z = -2 \\ y - 4z = 4 \end{cases}$$

Solution.

Recall that $x = A^{-1}b$,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 & 5/3 & 1/3 \\ -2 & 4/3 & -1/3 \\ -1/2 & 1/3 & -1/3 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4 \\ -6 \\ -5/2 \end{bmatrix}$$

Remark 3.5. $A \in M_n(\mathbb{R})$ is nonsingular iff A is row equivalent to I_n . And A is singular iff A is row equivalent to a matrix of zero row.

Definition 3.3. A matrix $E \in M_n(\mathbb{R})$ is called an elementary matrix if it can be obtained by applying a single elementary row operation on I_n .

Types of Elementary Matrices

Type 1. We denote the elementary matrix obtained when we switch rows k and ℓ of I_n as $P_{k\ell}$

$$\text{e.g. } \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = P_{14}; \quad P_{14}^{-1} = P_{14}$$

Type 2. We denote the elementary matrix obtained when we multiply row k of I_n by the nonzero constant c as $Q_k(c)$

$$\text{e.g. } \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = Q_3(c); \quad Q_3(c)^{-1} = Q_3\left(\frac{1}{c}\right)$$

Type 3. We denote the elementary matrix obtained when we add d times row k of I_n to row ℓ

$$\text{e.g. } \begin{bmatrix} 1 & d & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = R_{21}(d); \quad R_{21}(d)^{-1} = R_{21}(-d)$$

Theorem 3.4. The elementary matrices are nonsingular and their inverses are also elementary.

Remark 3.6. Some notable observations for elementary matrices are shown here:

1. The inverses of the elementary matrices can be simplified as follows:

- $P_{k\ell}^{-1} = P_{k\ell}$
- $Q_k(c)^{-1} = Q_k\left(\frac{1}{c}\right), c \neq 0$
- $R_{k\ell}(c)^{-1} = R_{k\ell}(-c)$

2. Let $E \in M_n(\mathbb{R})$ be an elementary matrix and $C \in M_{n \times p}(\mathbb{R})$. Then, we can denote the matrix obtained by applying C to the same row operation applied on I_n to get E as EC

Example 3.3. Let E_1, E_2, E_3 , and A be matrices defined as follows:

$$\begin{aligned} E_1 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, & E_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}, \\ E_3 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, & A &= \begin{bmatrix} 1 & \pi & 3 & 2 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & -\frac{3}{4} & -1 & 5 \end{bmatrix} \end{aligned}$$

Determine the elementary row operations corresponding to E_1, E_2 , and E_3 .

Solution. The elementary row operations represented by the given matrices are:

$$\begin{aligned} E_1 &\implies R_2 \leftarrow \sqrt{2}R_2 && (\text{or } Q_2(\sqrt{2})) \\ E_2 &\implies R_2 \leftarrow R_2 + 4R_3 && (\text{or } R_{32}(4)) \\ E_3 &\implies R_2 \longleftrightarrow R_3 && (\text{or } P_{23}) \end{aligned}$$

Example 3.4. Find $E_3E_2E_1A$ using the matrices from [Example 3.3](#).

Solution.

$$\begin{aligned} E_3E_2E_1A &= E_3E_2E_1 \begin{bmatrix} 1 & \pi & 3 & 2 \\ 0 & -\sqrt{2} & 0 & 0 \\ 0 & -3/4 & -1 & 5 \end{bmatrix} \\ &= E_3E_2 \begin{bmatrix} 1 & \pi & 3 & 2 \\ 0 & -2 & 0 & 0 \\ 0 & -3/4 & -1 & 5 \end{bmatrix} \\ &= E_3 \begin{bmatrix} 1 & \pi & 3 & 2 \\ 0 & -5 & -4 & 20 \\ 0 & -3/4 & -1 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 1 & \pi & 3 & 2 \\ 0 & -3/4 & -1 & 5 \\ 0 & -5 & -4 & 20 \end{bmatrix} \end{aligned}$$

Theorem 3.5. Let $C, D \in M_{m \times n}(\mathbb{R})$. Then C and D are row equivalent iff there exists elementary matrices $E_1, E_2, \dots, E_n \in M_n(\mathbb{R})$ such that

$$D = E_k \cdots E_2 E_1 C$$

Remark 3.7. $A \in M_n(\mathbb{R})$ is nonsingular iff:

- A is row equivalent to I_n
- $I_n = E_k \cdots E_2 E_1 A$ for some elementary $E_k, \dots, E_2, E_1 \in M_n(\mathbb{R})$
- $A^{-1} = E_k \cdots E_2 E_1$ and $A = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$ for some elementary $E_1, E_2, \dots, E_k \in M_n(\mathbb{R})$
- A is a product of elementary matrices

Example 3.5. Let $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$. Write A and A^{-1} as products of elementary matrices.

Solution. Let us first reduce A into the identity matrix and then identify the EROs used,

$$\begin{aligned} \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix} &\xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \\ &\xrightarrow{R_2 \rightarrow -1/2 R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \end{aligned}$$

Then,

$$I_2 = Q_2 \left(-\frac{1}{2} \right) R_{12}(-3) P_{12} A$$

Its inverse A^{-1} is:

$$\begin{aligned} A^{-1} &= Q_2 \left(-\frac{1}{2} \right) R_{12}(-3) P_{12} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

Recall that $A = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$, thus,

$$A = \begin{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \end{bmatrix}$$

Theorem 3.6. Let $A \in M_n(\mathbb{R})$. The following statements are equivalent:

- A is nonsingular
- A is row equivalent to I_n
- A is the product of elementary matrices
- The matrix equation $Ax = b$ has a unique solution for any $b \in M_{n \times 1}(\mathbb{R})$
- The homogeneous matrix equation $Ax = 0_{n \times 1}$, has only the **trivial solution** $x = 0_{n \times 1}$

Example 3.6. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is singular, since $Ax = 0_{3 \times 1}$ has a nontrivial solution namely

$$x = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}.$$

4 Determinant of a Square Matrix

Let $A = [a_{ij}] \in M_n(\mathbb{R})$. The determinant of A is defined as follows:

- If $n = 1$, then $\det A = a_{11}$
- Suppose $n \geq 1$ and assume that the determinant of a matrix of size $n - 1$ is defined. Let M_{ij} be the matrix obtained from A by deleting row i and column j of A . Define

$$C_{ij} = (-1)^{i+j} \det M_{ij}$$

wherein (i, j) is the cofactor of A .

Then we define the determinant as a cofactor expansion along row i ,

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

Similarly, we could also define it as a cofactor expansion along column j ,

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

Example 4.1. Obtain a formula for the determinant of $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

Solution. By cofactor expansion along row 1,

$$\begin{aligned} \det A &= a_{11}C_{11} + a_{12}C_{12} \\ &= a_{11}(-1)^{1+1} \det[a_{22}] + a_{12}(-1)^{1+2} \det[a_{21}] \\ &= \boxed{a_{11}a_{22} - a_{12}a_{21}} \end{aligned}$$

Example 4.2. Find the determinant of $B = \begin{bmatrix} 5 & 6 \\ 4 & 8 \end{bmatrix}$.

Solution. Recall that from the previous example, we have obtained a determinant formula for a square matrix of size 2, we could use that to solve for $\det B$

$$\det B = 5(8) - 6(4) = \boxed{16}$$

Example 4.3. Find the determinant of $B = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$.

Solution. By cofactor expansion along column 1,

$$\begin{aligned} \det B &= 3(-1)^{1+1} \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} + 2(-1)^{2+1} \begin{vmatrix} 1 & -4 \\ 4 & 8 \end{vmatrix} + 1(-1)^{3+1} \begin{vmatrix} 1 & -4 \\ 5 & 6 \end{vmatrix} \\ &= 3(16) - 2(24) + 26 = \boxed{26} \end{aligned}$$

Example 4.4. Find the determinant of $C = \begin{bmatrix} 2 & 3 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 4 & 6 & 1 & -5 \\ 0 & 5 & 3 & 0 \end{bmatrix}$.

Solution. Notice that column 4 has 3 zeros which could help us compute for the determinant for efficiently,

$$\det C = 0 + 0 - 5(-1)^{3+4} \begin{vmatrix} 2 & 3 & 0 \\ 1 & 1 & 0 \\ 0 & 5 & 3 \end{vmatrix} + 0 = 5 \begin{vmatrix} 2 & 3 & 0 \\ 1 & 1 & 0 \\ 0 & 5 & 3 \end{vmatrix}$$

We will perform cofactor expansion along column 3 since it has the least amount of zeros,

$$= 5 \left(3(-1)^{3+3} \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} \right) = 15(2 - 3) = \boxed{-15}$$

Example 4.5. The examples that will be shown here will help us determine some properties of determinants. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, and $B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$

- Suppose we switch row 1 and row 2 of matrix A , then the determinant is

$$\begin{vmatrix} c & d \\ a & b \end{vmatrix} = bc - ad = -(ad - bc) = -\det A$$

- Suppose we add a constant $r \in \mathbb{R}$ to row 2 of matrix A , then the determinant is

$$\begin{vmatrix} a & b \\ rc & rd \end{vmatrix} = ar d - br c = r(ad - bc) = r \det A$$

- Suppose we multiply r to row 1 of matrix A and then add it to row 2, then the determinant is

$$\begin{vmatrix} a & b \\ ra + c & rb + d \end{vmatrix} = a(rb + d) - b(ra + c) = ad - bc = \det A$$

- Suppose we perform cofactor expansion along row 3 of matrix B to get its determinant,

$$\det B = a_{33}(-1)^{3+3} \begin{vmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{vmatrix} = a_{33}(a_{11}a_{22}) = a_{11}a_{22}a_{33}$$

Some Properties of Determinant. Let $A, B \in M_n(\mathbb{R})$,

1. If B is obtained from A by:

- switching two rows/columns of A then $\det B = -\det A$
- multiplying $r \in \mathbb{R}$ to a row/columns of A then $\det B = r \det A$

$$\text{e.g. } \det(rA) = \underbrace{r \times r \times \cdots \times r}_n \det A = r^n \det A$$

- adding a multiple of a row/column of A to another row/column of A then $\det B = \det A$

2. $\det(A^T) = \det A$

3. If A has a row/column full of 0s then $\det A = 0$.

4. If A has two equal rows/columns, then $\det A = 0$.

5. If $A = [a_{ij}]$ is upper or lower triangular, then $\det A = a_{11}a_{22} \cdots a_{nn}$, which is the product of all the main diagonal entries of the matrix.
6. $\det(AB) = \det A \det B$, and in particular, $\det(A^k) = (\det A)^k$ for all $k \in \mathbb{Z}^+$.

Theorem 4.1. If $E \in M_n(\mathbb{R})$ is elementary, then $\det E \neq 0$.

Theorem 4.2. $A \in M_n(\mathbb{R})$ is nonsingular iff $\det A \neq 0$, in this case

$$\det(A^{-1}) = \frac{1}{\det A}$$

Definition 4.1. Let $A \in M_n(\mathbb{R})$ and C_{ij} be the (i, j) column of A . The **adjoint** of A is $\text{adj } A = [C_{ij}]^T$.

Example 4.6. Find the adjoint of the following:

1. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Solution.

$$\begin{aligned} C_{11} &= (-1)^{1+1}d = d & C_{21} &= (-1)^{2+1}b = -b \\ C_{12} &= (-1)^{1+2}c = -c & C_{22} &= (-1)^{2+2}a = a \end{aligned}$$

$$\therefore \text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

2. $B = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 4 \\ 5 & 8 & 9 \end{bmatrix}$

Solution.

$$\begin{aligned} C_{11} &= \begin{vmatrix} 3 & 4 \\ 8 & 9 \end{vmatrix} = -5 & C_{21} &= -\begin{vmatrix} 1 & 0 \\ 8 & 9 \end{vmatrix} = -9 & C_{31} &= \begin{vmatrix} 1 & 0 \\ 3 & 4 \end{vmatrix} = 4 \\ C_{12} &= -\begin{vmatrix} 2 & 4 \\ 5 & 9 \end{vmatrix} = 2 & C_{22} &= \begin{vmatrix} 1 & 0 \\ 5 & 9 \end{vmatrix} = 9 & C_{32} &= -\begin{vmatrix} 1 & 0 \\ 2 & 4 \end{vmatrix} = -4 \\ C_{13} &= \begin{vmatrix} 2 & 3 \\ 5 & 8 \end{vmatrix} = 1 & C_{23} &= -\begin{vmatrix} 1 & 1 \\ 5 & 8 \end{vmatrix} = -3 & C_{33} &= \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 1 \end{aligned}$$

$$\therefore \text{adj } B = \begin{bmatrix} -5 & -9 & 4 \\ 2 & 9 & -4 \\ 1 & -3 & 1 \end{bmatrix}$$

Theorem 4.3 (Adjoint Formula). If $A \in M_n(\mathbb{R})$ then,

$$A(\text{adj } A) = (\det A)I_n = (\text{adj } A)A$$

If A is nonsingular then,

$$\begin{aligned} A \left(\frac{1}{\det A} \cdot \text{adj } A \right) &= I_n \\ A^{-1} &= \frac{1}{\det A} \text{adj } A \end{aligned}$$

Example 4.7. If $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is nonsingular then,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 4.8. Find the inverse of $B = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 4 \\ 5 & 8 & 9 \end{bmatrix}$.

Solution.

Recall that if we perform Type III operations in any row or column of a matrix, then its determinant does not change. Thus, we could multiply -1 to the the first column and add that to the second column, we would get the following matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 4 \\ 5 & 3 & 9 \end{bmatrix} \implies \det B = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 4 \\ 5 & 3 & 9 \end{vmatrix}$$

Let us perform cofactor expansion along row 1,

$$\det B = 1 \begin{vmatrix} 1 & 4 \\ 3 & 9 \end{vmatrix} = 9 - 12 = -3$$

Since we have already obtained $\text{adj } B$ from [Example 4.6](#), we could use that to find the inverse using the adjoint formula,

$$B^{-1} = -\frac{1}{3} \begin{bmatrix} -5 & -9 & 4 \\ 2 & 9 & -4 \\ 1 & -3 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 5/3 & 3 & -4/3 \\ -2/3 & -3 & 4/3 \\ -1/3 & 1 & -1/3 \end{bmatrix}}$$

Theorem 4.4 (Cramer's Rule). Let $A \in M_n(\mathbb{R})$ be nonsingular and $b \in M_{n \times 1}(\mathbb{R})$. The matrix equation $Ax = b$ has a unique solution $x = [x_1 \ x_2 \ \cdots \ x_n]^T$ given by

$$x_i = \frac{\det A_i}{\det A}$$

where A_i is the matrix obtained by replacing column i of A by b .

Example 4.9. Solve for y only in the system of equations below

$$\begin{cases} x + y + z = 8 \\ -x - 2y + 3z = 1 \\ 3x - 7y + 4z = 0 \end{cases}$$

Solution.

Let A be the coefficient matrix and b be the column vector of the constants,

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & 3 \\ 3 & -7 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 8 \\ 1 \\ 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 8 & 1 \\ -1 & 1 & 3 \\ 3 & 0 & 4 \end{bmatrix}$$

By Cramer's Rule,

$$y = \frac{\det A_2}{\det A} = \frac{105}{39} = \boxed{\frac{35}{13}}$$

5 Vector Spaces and Subspaces

Definition 5.1. A **vector space** (over $\mathbb{R}$) is a set V equipped with two operations: $\oplus$ (vector addition) and $\odot$ (scalar multiplication) satisfying the following for all $u, v, w \in V$ and for all $c, d \in \mathbb{R}$:

- A1. $u \oplus v \in V$ (closure)
- A2. $(u \oplus v) \oplus w = u \oplus (v \oplus w)$ (associativity)
- A3. $u \oplus v = v \oplus u$ (commutativity)
- A4. There exists a zero vector $\vec{0} \in V$ such that $u \oplus \vec{0} = u$
- A4. For any $u \in V$, there exists an inverse $-u \in V$ such that $u \oplus (-u) = \vec{0}$

- S1. $c \odot u \in V$ (closure)
- S2. $(c + d) \odot u = (c \odot u) \oplus (d \odot u)$
- S3. $c \odot (u \oplus v) = (c \odot u) \oplus (c \odot v)$
- S4. $(cd) \odot u = c \odot (d \odot u)$
- S5. $1 \odot u = u$

Remark 5.1. Let V be a vector space

1. The elements of V are called **vectors**, and the set of all real numbers are **scalars**.
2. There are 2 different:
 - (a) additions: $\oplus$ on V and $+$ on $\mathbb{R}$
 - (b) multiplications: $\odot$ on V and $\cdot$ on $\mathbb{R}$
 - (c) zeros: $\vec{0}$ (zero vector) and 0 (zero scalar)

If the context is clear we will use the $+$ and $\cdot$ in place of $\oplus$ or $\odot$.

Example 5.1. The following are vector spaces:

1. $\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) \mid a_1, a_2, \dots, a_n \in \mathbb{R}\}$ under

$$(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$c(a_1, a_2, \dots, a_n) = (ca_1, ca_2, \dots, ca_n)$$

Wherein the zero vector $\vec{0} = (\underbrace{0, 0, \dots, 0}_n)$ and its inverse is $-(a_1, a_2, \dots, a_n) = (-a_1, -a_2, \dots, -a_n)$

2. $M_{m \times n}(\mathbb{R})$ under matrix addition and scalar multiplication on a matrix
3. $P_n = \{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \mid a_0, a_1, \dots, a_n \in \mathbb{R}\}$
4. $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R} \wedge i^2 = -1\}$
5. $\mathcal{F}(\mathbb{R})$ which is the set of all real-valued functions with domain $\mathbb{R}$ under

$$(f + g)(x) = f(x) + g(x)$$

$$(cf)(x) = cf(x), \quad c \in \mathbb{R}$$

6. $V = (0, +\infty) = \mathbb{R}^+$ under

$$x \oplus y = xy \in V$$

$$r \odot x = x^r \in V$$

Theorem 5.1. Let V be a vector space, $u \in V$ and $c \in \mathbb{R}$. Then

- $\vec{0}$ is unique and the additive inverse $-u$ is unique.
- $c\vec{0} = c(\vec{0} + \vec{0})$
- $cu = \vec{0} \implies c = 0$ or $u = \vec{0}$
- $-(cu) = (-c)u = c(-u)$
- $-u = (-1)u$

Definition 5.2. Let V be a vector space and $W \subseteq V$. If W is also a vector space under the same operations as V , then W is called a **subspace** of V , denoted by $W \leq V$.

Example 5.2.

1. If V is a vector space then the following are subspaces of V :
 - (a) V (improper subspace)
 - (b) $\{\vec{0}\}$ (trivial subspace)
2. $P_1 = \{ax + b \mid a, b \in \mathbb{R}\}$ is a subspace of $P_2 = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}$.

Theorem 5.2 (Subspace Test). Let V be a vector space and $W \subseteq V$. Then $W \leq V$ iff the following are satisfied:

- $W \neq \emptyset$
- For all $w_1, w_2 \in W$, $w_1 + w_2 \in W$
- For all $c \in \mathbb{R}$, and for all $w \in W$, $cw \in W$

Remark 5.2. A nonempty subset W of V is NOT a subspace of V iff at least one of the following is true:

1. There exists $w_1, w_2 \in W$ such that $w_1 + w_2 \notin W$
2. There exists $c \in \mathbb{R}$ such that $cw \notin W$

Example 5.3. Let $V = \mathbb{R}^3$. Show that $W = \{(x, y, z) \mid x + y + z = 0\}$ is a subspace of $\mathbb{R}^3$.

Proof.

- $(0, 0, 0) \in W$ since $0 + 0 + 0 = 0$, thus, $W \neq \emptyset$.
- Let $w_1 = (x_1, y_1, z_1), w_2 = (x_2, y_2, z_2) \in W$

$$w_1 + w_2 = (x_1 + x_2, y_1 + y_2, z_1 + z_2) \in W$$

$$\begin{aligned} (x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) &= (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) \\ &= 0 \end{aligned}$$

Thus, $w_1 + w_2 \in W$

- Let $c \in \mathbb{R}$ and $w = (x, y, z) \in W$

$$c(x, y, z) = (cx, cy, cz)$$

$$c(x + y + z) = c(0)$$

Thus, $cw \in W$

Therefore, $W \leq \mathbb{R}^3$ by the Subspace Test. ■

Example 5.4. Let $V = M_2(\mathbb{R})$. Show that

$$W = \{A \in M_2(\mathbb{R}) \mid A \text{ is upper triangular}\}$$

is a subspace of V .

Proof.

- $0_{2 \times 2} \in W \implies W \neq \emptyset$
- Let $A = \begin{bmatrix} a_1 & a_2 \\ 0 & a_3 \end{bmatrix}$, $B = \begin{bmatrix} b_1 & b_2 \\ 0 & b_3 \end{bmatrix}$, and $c \in \mathbb{R}$,

$$A + B = \begin{bmatrix} a_1 + b_1 & a_2 + b_2 \\ 0 & a_3 + b_3 \end{bmatrix} \in W$$

$$cA = \begin{bmatrix} ca_1 & ca_2 \\ 0 & ca_3 \end{bmatrix} \in W$$

Therefore, $W \leq M_2(\mathbb{R})$ by the Subspace Test. ■

Example 5.5. Let $V = M_2(\mathbb{R})$. Show that

$$W = \{A \in M_2(\mathbb{R}) \mid A \text{ is singular}\}$$

is **NOT** a subspace of V .

Proof. By counter example, let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$,

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \notin W$$

Therefore, $W \not\leq V$ by the Subspace Test. ■

Example 5.6. Let $V = P_1 = \{ax + b \mid a, b \in \mathbb{R}\}$. Prove whether the following are subspaces of P_1 or not:

- a. $W = \{ax - 2a \mid a \in \mathbb{R}\}$

Proof.

- $0 = 0x = 0 \in W \rightarrow W \neq \emptyset$
- Let $f = ax - 2a$, $g = bx - 2b$, and $c \in \mathbb{R}$. Then

$$f + g = (a + b)x - 2(a + b) \in W$$

$$cf = c(ax - 2a) = (ac)x - 2(ac) \in W$$

Therefore, $W \leq P_1$, by the Subspace Test. ■

- b. $W = \{ax + b \mid a + b = 1\}$

Proof. Take $f = x$ and $c = \frac{1}{2}$,

$$cf = \frac{1}{2}x \notin W$$

Therefore, $W \not\leq P_1$, by the Subspace Test. ■

6 Spanning Sets and Linear Independence

Consider a vector space $V = \mathbb{R}^3$ and three vectors $\hat{i}, \hat{j}$, and $\hat{k}$ where $\hat{i} = (1, 0, 0), \hat{j} = (0, 1, 0), \hat{k} = (0, 0, 1) \in \mathbb{R}^3$. Then every vector $(a, b, c) \in \mathbb{R}^3$ can be expressed as

$$\begin{aligned}(a, b, c) &= (a, 0, 0) + (0, b, 0) + (0, 0, c) \\ &= a\hat{i} + b\hat{j} + c\hat{k}\end{aligned}$$

Every vector in $\mathbb{R}^3$ can be expressed in terms of $\hat{i}, \hat{j}$, and $\hat{k}$.

Definition 6.1. Let V be a vector space and $v_1, v_2, \dots, v_k \in V$. A **linear combination** of $v_1, v_2, \dots, v_k$ is any element of the form

$$a_1v_1 + a_2v_2 + \dots + a_kv_k = \sum_{i=1}^k a_iv_i$$

where $a_i \in \mathbb{R}$ for all i .

Example 6.1. The following are examples of linear combinations:

1. $0v_1 + 0v_2 + \dots + 0v_k = \vec{0}$ is a linear combination of $v_1, v_2, \dots, v_k$.
2. Every vector in $\mathbb{R}^3$ is a linear combination of $\hat{i}, \hat{j}, \hat{k}$.
3. In the vector space P_4 , $4x^4 - 2x^3 + 6x - 9 = 4x^4 + (-2)x^3 + 0x^2 + 6x + (-9)(1)$ is a linear combination of $1, x, x^2, x^3, x^4$.
4. Let $A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 2 & 2 \end{bmatrix}$ and $x = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Then

$$\begin{aligned}Ax &= \begin{bmatrix} 1 & -3 & 0 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a - 3b \\ 2b + 2c \end{bmatrix} \\ &= \begin{bmatrix} a \\ 0 \end{bmatrix} + \begin{bmatrix} -3b \\ 2b \end{bmatrix} + \begin{bmatrix} 0 \\ c \end{bmatrix} \\ &= a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -3 \\ 2 \end{bmatrix} + c \begin{bmatrix} 0 \\ 2 \end{bmatrix}\end{aligned}$$

Ax is a linear combination of the columns of A . In general if $A \in M_{m \times n}(\mathbb{R})$ and $x \in M_{n \times 1}(\mathbb{R})$, then Ax is a linear combination of the columns of A .

5. Let $v = (-3, -\frac{1}{2}, \frac{5}{2})$. Is v a linear combination of $(1, 0, 0)$, $(2, 1, 0)$, and $(0, 1, 1)$?

Solution. We want to know if there exists scalars a, b, c such that

$$\begin{aligned}\left(-3, -\frac{1}{2}, \frac{5}{2}\right) &= a(1, 0, 0) + b(2, 1, 0) + c(0, 1, 1) \\ &= (a + 2b, b + c, c)\end{aligned}$$

Comparing coordinates,

6. In $V = \mathcal{F}(\mathbb{R})$, $\cosh \in V$ is a linear combination of e^x and e^{-x}

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{1}{2}e^x + \frac{1}{2}e^{-x}$$

Definition 6.2. Let $S = \{v_1, v_2, \dots, v_k\}$ be a nonempty subset of a vector space V

- The span of S , denoted by $\text{span } S$, is the set of all linear combinations of elements of S , that is,

$$\text{span } S = \{a_1 v_1 + a_2 v_2 + \dots + a_k v_k \mid a_i \in \mathbb{R} \forall i\}$$

- If $\text{span } S = V$, we say that S is a spanning set of V or V is spanned by S .⁶

Example 6.2. Let $V = \mathbb{R}^2$ and $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$, then,

$$\begin{aligned} \text{span } S &= \{a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1) \mid a, b, c \in \mathbb{R}\} \\ &= \{(a, b, c) \mid a, b, c \in \mathbb{R}\} \end{aligned}$$

Therefore, S spans $\mathbb{R}^3$.

Example 6.3. Let $V = P_n$ and $S = \{1, x, x^2, \dots, x^n\}$, then,

$$\text{span } S = \{a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n \mid a_i \in \mathbb{R} \forall i\} = P_n$$

Therefore, S spans P_n .

Example 6.4. Let $V = \mathbb{C} = \{a + bi \mid a, b \in \mathbb{R} \text{ and } i^2 = -1\}$ and $S = \{1, i\}$, then,

$$\text{span } S = \{a + bi \mid a, b \in \mathbb{R} \text{ and } i^2 = -1\} = \mathbb{C}$$

Therefore, S spans $\mathbb{C}$.

Example 6.5. Let $V = M_2(\mathbb{R})$ and $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$, then,

$$\text{span } S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{R} \right\} = M_2(\mathbb{R})$$

Therefore S spans $M_2(\mathbb{R})$.

Example 6.6. Let $S = \{(2, 3), (1, 2)\}$. Does this span $\mathbb{R}^2$?

Solution. Let $(a, b) \in \mathbb{R}^2$. We want to know if there exists $x, y \in \mathbb{R}$ such that,

$$\begin{aligned} (a, b) &= x(2, 3) + y(1, 2) \\ &= (2x, 3x) + (y, 2y) \\ &= (2x + y, 3x + 2y) \end{aligned}$$

$$\begin{cases} 2x + y = a \\ 3x + 2y = b \end{cases} \iff \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

The coefficient matrix A of the system is nonsingular, since $\det A \neq 0$, therefore the system has a solution.

$$\begin{aligned} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}^{-1} \begin{bmatrix} a \\ b \end{bmatrix} \\ &= \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2a - b \\ 3a - 2b \end{bmatrix} \end{aligned}$$

Therefore, S spans $\mathbb{R}^2$.

⁶Every vector in V is a linear combination of elements in S

Example 6.7. Let $S = \{x^2 + 2x + 1, x^2 + 2\} \subseteq P_2$. Does S span P_2 ?

Solution. Let $ax^2 + bx + c \in P_2$. We want to know if there exists $D, E \in \mathbb{R}$ such that

$$\begin{aligned} ax^2 + bx + c &= D(x^2 + 2x + 1) + E(x^2 + 2) \\ &= x^2(D + E) + x(2D) + (D + 2E) \end{aligned}$$

$$\begin{cases} D + E = a \\ 2D = b \\ D + 2E = c \end{cases} \longleftrightarrow \left[\begin{array}{cc|c} 1 & 0 & a \\ 2 & 0 & b \\ 1 & 2 & c \end{array} \right] \xrightarrow{EROs} \left[\begin{array}{cc|c} 1 & 1 & a \\ 0 & 1 & -a + c \\ 0 & 0 & -4a + b + 2c \end{array} \right]$$

Since there is a restriction placed on the last row, such that $-4a + b + 2c$ must be equal to 0, therefore S does not span P_2 .

Theorem 6.1. If $S = \{v_1, v_2, \dots, v_k\}$ is a nonempty subset of V , then $\text{span } S \leq V$.

Remark 6.1. Shown below are some remarks regarding the span of sets:

- If $S \subseteq \text{span } S$ and $\text{span } S$ is the smallest subspace of V containing S .
- $\text{span}(\emptyset) = \{\vec{0}\}$

Definition 6.3. Let $S = \{v_1, v_2, \dots, v_n\}$ be a nonempty subset of a vector space V . Then S is said to be **linearly independent** if whenever

$$a_1v_1 + a_2v_2 + \dots + a_nv_n = \vec{0}$$

then $a_1 = a_2 = \dots = a_n = 0$. Otherwise, S is said to be **linearly dependent**.

Remark 6.2. The following are some remarks on linear dependence:

1. S is linearly dependent iff there exists $a_1, a_2, \dots, a_n \in \mathbb{R}$ and not all zero, such that,

$$\sum_{i=1}^n a_i v_i = \vec{0}$$

2. $\emptyset$ is taken to be linearly independent.

Example 6.8. Determine if the set $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \subseteq \mathbb{R}^3$ is linearly independent.

Solution. Form the equation,

$$\begin{aligned} a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1) &= (0, 0, 0) \\ (a, b, c) &= (0, 0, 0) \end{aligned}$$

Because $a = b = c = 0$, therefore S is linearly independent.

In general, $S = \{e_1, e_2, \dots, e_n\}$ is a linear independent subset of $\mathbb{R}^n$.

Example 6.9. Determine whether the set $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \subseteq M_2(\mathbb{R})$ is linearly independent.

Solution. Form the equation,

$$a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\
&= \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}
\end{aligned}$$

where $a, b, c, d \in \mathbb{R}$. Since $a = b = c = d = 0$, therefore, S is linearly independent.

In general, $S = \{E_{k\ell} \mid k = 1, 2, \dots, m, \ell = 1, 2, \dots, n\}$ is a linear independent subset of $M_{m \times n}(\mathbb{R})$. As an exercise, determine if the set S is a spanning set in [Example 6.5](#) and [Example 6.6](#).

Example 6.10. Determine if the set $S = \{e^x, e^{-x}\} \subseteq \mathcal{F}(\mathbb{R})$.

Solution. Form the equation,

$$ae^x + be^{-x} = 0$$

where $a, b \in \mathbb{R}$. By differentiating both sides we obtain,

$$ae^x - be^{-x} = 0$$

Adding the first and second equation together,

$$ae^x + be^{-x} + ae^x - be^{-x} = 0$$

$$2ae^x = 0 \implies a = 0$$

Since $a = b = 0$, therefore, S is linearly independent.

Remark 6.3. Let V be a vector space and $S, T \leq V$ be finite:

1. If $\vec{0} \in S$, then S is linearly dependent.
2. If $\vec{0} + v \in V$, then $\{v\}$ is linearly independent.
3. Suppose $S \subseteq T$
 - (a) $\text{span } S \subseteq \text{span } T$
 - (b) If T is linearly independent then S is linearly independent
 - (c) If S is linearly dependent then T is linearly dependent
4. S is linearly dependent iff some vector in S is a linear combination of the other vectors in S . In particular, $S = \{u, v\} \quad (u, v \neq 0)$ is linearly dependent iff $v = cu$ for some $0 \neq c \in \mathbb{R}$. In this case, u and v are said to be parallel.

7 Basis and Dimension of a Vector Space

Definition 7.1. A subset B of a vector space V is called a basis for V if $\text{span } B = V$ and B is linearly independent.

Example 7.1.

1. $\emptyset$ is a basis for $\{\vec{0}\}$
2. $B = \{e_1, e_2, \dots, e_n\} \subseteq \mathbb{R}^n$ is a basis for $\mathbb{R}^n$
3. $B = \{E_{kl} \mid k = 1, 2, \dots, m \quad l = 1, 2, \dots, n\}$ is a basis for $M_{m \times n}(\mathbb{R})$
4. $B = \{1, x, x^2, \dots, x^n\}$ is a basis for P_n
5. $B = \{1, i\}$ is a basis for $\mathbb{C}$.

Remark 7.1. The bases in Example 2 - 5 as shown above are known as the **standard bases** of their respective spaces.

Example 7.2. Is $B = \{(2, 3), (1, 2)\}$ a basis for $\mathbb{R}^2$?

Solution. Recall that a set B is a basis if $\text{span } B = V$ and B is linearly independent. Let us verify whether these conditions hold true:

$$x(2, 3) + y(1, 2) = (a, b)$$

$$(2x + y, 3x + 2y) = (a, b)$$

To verify, whether B is linearly independent,

$$(2x + y, 3x + 2y) = (0, 0)$$

$$\left[\begin{array}{cc|c} 2 & 1 & 0 \\ 3 & 2 & 0 \end{array} \right] \xrightarrow{EROs} \left[\begin{array}{cc|c} 2 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right]$$

Because $\det B \neq 0$ it is nonsingular and its solution is the trivial solution. Therefore, B is linearly independent and is a basis for $\mathbb{R}^2$.

Example 7.3. Is $B = \{x^2 + x, x^2 + 1, 1\}$ a basis for P_2 ?

Solution. Let us first verify if B is a spanning set. Let $ax^2 + bx + c \in P_2$, and $D, E, F \in \mathbb{R}$,

$$ax^2 + bx + c = D(x^2 + x) + E(x^2 + 1) + F(1)$$

$$= x^2(D + E) + x(D) + (E + F)$$

$$\begin{cases} D + E = a \\ D = b \\ E + F = c \end{cases} \longleftrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & a \\ 1 & 0 & 0 & b \\ 0 & 1 & 1 & c \end{array} \right]$$

Since the system of equations has a solution with the form $(a, a - b, c - a + b)$. Therefore, it spans P_2 . To verify that it is linearly independent,

$$0 = x^2(D + E) + x(D) + (E + F)$$

Since the system of equations forces the other variables to be 0, thus, B is linearly independent. Therefore, B is a basis for P_2 .⁷

⁷An alternative solution would be to use determinants.

Theorem 7.1. If $B = \{v_1, v_2, \dots, v_k\}$ is a basis for a vector space V , then every vector in V can be expressed in exactly one way as a linear combination of vectors in B .

$$v = a_1v_1 + a_2v_2 + \dots + a_kv_k$$

Theorem 7.2. Let V be a vector space having a finite spanning set S . Then S has a subset B that is a basis for V . To find a basis for V with finite spanning set $S = \{v_1, v_2, \dots, v_k\}$

1. Form $\vec{0} = x_1v_1 + x_2v_2 + \dots + x_kv_k$ where each $x_i \in \mathbb{R}$ is unknown.
2. Use EROs to bring the coefficient matrix to REF or RREF.
3. The vectors corresponding to columns containing the leading entries forms a basis for V .

Example 7.4. Find a basis for the given subspace

$$W := \text{span}\{x^2, x + 4, 2x + 7, 1\} \leq P_2$$

Solution.

A spanning set for W is $S = \{x^2, x + 4, 2x + 7, 1\}$

$$\vec{0} = Ax^2 + B(x + 4) + C(2x + 7) + D$$

$$\vec{0} = x^2(A) + x(B + 2C) + (4B + 7C + D)$$

Forming the augmented matrix,

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 4 & 7 & 1 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right]$$

And since the fourth column has no leading entries, we can remove it. Thus, $B = \{x^2, x + 4, 2x + 7\}$ is a basis for W .

Example 7.5. Find a basis for the given subspace $W = \text{span}\{(1, -1, 3, 2), (0, -1, 2, 1), (2, 1, 0, 1)\} \subseteq \mathbb{R}^4$.

Solution.

$$a(1, -1, 3, 2) + b(0, -1, 2, 1) + c(2, 1, 0, 1) = (0, 0, 0, 0)$$

$$(a + 2c, -a - b + c, 3a + 2b, 2a + b + c) = (0, 0, 0, 0)$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ -1 & -1 & 1 & 0 & 0 \\ 3 & 2 & 0 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Therefore, $\{(1, -1, 3, 2), (0, -1, 2, 1)\}$ is a basis for W .

Example 7.6. Find a basis for the given subspace $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_2(\mathbb{R}) \mid b + 2c - d = 0 \right\}$.

Solution.

Let us first find a spanning set,

$$\begin{aligned}
 A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in W &\longleftrightarrow b + 2c - d = 0 \\
 &\longleftrightarrow \begin{bmatrix} a & b \\ c & b + 2c \end{bmatrix} \\
 &\longleftrightarrow \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & b \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & 2c \end{bmatrix} \\
 &\longleftrightarrow a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} \\
 W = \text{span} \underbrace{\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} \right\}}_B
 \end{aligned}$$

Thus, B spans W . Form,

$$\begin{aligned}
 a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad a, b, c \in \mathbb{R} \\
 \begin{bmatrix} a & b \\ c & b + 2c \end{bmatrix} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\
 \implies \begin{cases} a = 0 \\ b = 0 \\ c = 0 \\ b + 2c = 0 \end{cases}
 \end{aligned}$$

And since $a = b = c = 0$, B is linearly independent. Therefore, B is a basis for W .

Example 7.7. Find a basis for $W = \{ax^3 + bx^2 + cx + d \in P_3 \mid a + c = 0 \text{ and } b + d = 0\}$.

Solution.

Let $f(x) = ax^3 + bx^2 + cx + d \in W$ and since there are restrictions, it must follow that $c = -a$ and $d = -b$,

$$\begin{aligned}
 ax^3 + bx^2 - ax - b &= a(x^3 - x) + b(x^2 - 1) \\
 f(x) &\in \text{span}\{x^3 - x, x^2 - 1\} \\
 \therefore W &= \text{span} \underbrace{\{x^3 - x, x^2 - 1\}}_B
 \end{aligned}$$

Thus, B spans W . And since $x^2 - 1$ is not a scalar multiple of $x^3 - x$, B is linearly independent. Therefore, B is a basis for W .

Definition 7.2. A vector space V with a finite spanning set is said to be **finite dimensional**. Otherwise, V is said to be **infinite dimensional**.

Remark 7.2. The following are some remarks on basis:

1. Every finite-dimensional vector space has a finite basis.
2. Every infinite-dimensional vector space has an infinite basis.

Example:

- The vector space $\mathbb{R}^n, M_{m \times n}(\mathbb{R}), P_n, \mathbb{C}$ are all finite-dimensional.
- $\mathcal{F}(\mathbb{R})$ is infinite-dimensional.
- $\mathbb{R}[x]$ the set of all polynomials in the variable x having real coefficients is an infinite-dimensional vector space under addition and scalar multiplication of polynomials. ($B = \{x^n \mid 0 \leq n \in \mathbb{Z}\}$ is an infinite basis for $\mathbb{R}[x]$).

Theorem 7.3 (Basis Completion Theorem). Let V be a finite-dimensional vector space and $S \subseteq V$ be a linearly independent subset of V . Then V has a basis B containing S .

To extend a linearly independent set S to a basis for V

1. Let C be a known basis (standard basis) for V . Then $T := S \cup C$ is a spanning set for V .
2. Use the preceding algorithm to get a subset B of T that is a basis for V , Then B contains S .

Example 7.8. Find a basis for $M_{3 \times 1}(\mathbb{R})$ containing $v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Solution. Verify that $S = \{v_1, v_2\}$ is linearly independent.

Let $C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ be the standard basis for $M_{3 \times 1}(\mathbb{R})$.

Form

$$T = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Using the previous algorithm, we obtain the following coefficient matrix,

$$\begin{bmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 1/3 & 1/3 & 0 \\ 0 & 1 & 1/3 & -2/3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Since there are no pivots for columns 3 and 4. We can delete them. Therefore, $B = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

is a basis for $M_{3 \times 1}(\mathbb{R})$ containing v_1 and v_2 .

Theorem 7.4 (Steinitz Replacement Theorem). Let V be a finite-dimensional vector space with finite spanning set S . If T is a linearly independent subset of V , then $|T| \leq |S|$.

Corollary 7.1. Any two bases of a finite-dimensional vector space V has the same number of elements.

Definition 7.3. The number of elements of any basis of a finite-dimensional vector space V is called the dimension of V , denoted by $\dim V$. If $n = \dim V$, then V is said to be n -dimensional.

Remark 7.3. If V is an infinite-dimensional vector space, then $\dim V = +\infty$.

8 The Fundamental Subspaces and Rank of a Matrix

Definition 8.1. Let $A \in M_{m \times n}(\mathbb{R})$ with rows $R_1, R_2, \dots, R_m$ and columns $C_1, C_2, \dots, C_n$.

1. The **row space** of A is $R(A) := \text{span}\{R_1, R_2, \dots, R_m\} \leq M_{1 \times n}(\mathbb{R})$
2. The **column space** of A is $C(A) := \text{span}\{C_1, C_2, \dots, C_n\} \leq M_{m \times 1}(\mathbb{R})$
3. The **row rank** (respectively **column rank**) of A is $\dim R(A)$ (respectively $\dim C(A)$)

To find bases for $R(A)$ and $C(A)$:

1. Use elementary row operations to bring A to a matrix U in REF.
2. Basis for $C(A)$ = columns of A corresponding to columns of U containing leading entries.
3. Basis for $R(A)$ = nonzero rows of the REF matrix U .

Theorem 8.1. The row rank and column rank of any $A \in M_{m \times n}(\mathbb{R})$ are equal.

Definition 8.2. The rank of $A \in M_{m \times n}(\mathbb{R})$, denoted by $\text{rank } A$, is the common value of the row rank and column ranks of A .

Definition 8.3. Let $A \in M_{m \times n}(\mathbb{R})$. The null space of A , denoted by $N(A)$, is the solution set of the system $Ax = 0_{m \times 1}$. That is,

$$N(A) = \{x \in M_{n \times 1}(\mathbb{R}) \mid Ax = 0_{m \times 1}\}$$

Remark 8.1. If $A \in M_{m \times n}(\mathbb{R})$, then $N(A) \leq M_{n \times 1}(\mathbb{R})$. The **nullity** of A , denoted by $v(A)$, is $\dim N(A)$.

To find $N(A)$, solve for x in the system $Ax = 0_{m \times 1}$.

Example 8.1. For each matrix A , find a basis for $N(A)$. Identify the nullity of A .

$$A = \begin{bmatrix} 1 & 0 \\ 4 & -2 \\ 2 & 3 \end{bmatrix}$$

Solution.

By reducing matrix A to RREF, we get

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$$

Therefore, $N(A) = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$ and $v(A) = 0$.

Example 8.2. Find a basis for the given system of equations

$$\begin{cases} x_1 + 2x_3 + x_5 = 0 \\ x_2 + x_3 + x_5 = 0 \\ x_4 - x_5 = 0 \end{cases}$$

Theorem 8.2 (Rank-Nullity Theorem). If $A \in M_{m \times n}(\mathbb{R})$, then $n = \text{rank } A + v(A)$,

9 Inner Product Spaces

Definition 9.1. An inner product on a vector space V is a function $\langle \cdot, \cdot \rangle$ that assigns two vectors $u, v \in V$ to a scalar $\langle u, v \rangle \in \mathbb{R}$ satisfying the following for all $u, v, w \in V$ and for all $c \in \mathbb{R}$

- IP1. $\langle u, v \rangle = \langle v, u \rangle$ (symmetric)
- IP2. $\langle u, u \rangle \geq 0$ (nonnegativity)
- IP3. $\langle u, u \rangle = 0$ iff $u = \vec{0}$ (definite)
- IP4. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ (additive on the first component)
- IP5. $\langle cu, w \rangle = c\langle u, w \rangle$ (homogeneous on the first component)

Remark 9.1. A vector space V with an inner product is called an inner product space.⁸

Definition 9.2. Let V be an inner product space and $u, v \in V$

- 1. The **norm/magnitude** of u is $\|u\| = \sqrt{\langle u, u \rangle}$
- 2. u is called the **unit vector** if $\|u\| = 1$
- 3. The **distance** between u and v is $d(u, v) = \|u - v\|$

Remark 9.2. Let V be an inner product space, $u \in V$, $c \in \mathbb{R}$

- 1. $\|u\|^2 = \langle u, u \rangle$
- 2. $\|u\| \geq 0$
- 3. $\|\vec{0}\| = \sqrt{\langle \vec{0}, \vec{0} \rangle} = 0$
- 4. $\|u\| = 0$ iff $u = \vec{0}$
- 5. $\|cu\| = \sqrt{\langle cu, cu \rangle} = |c|\|u\|$
- 6. If $u \neq \vec{0}$, then $\frac{u}{\|u\|}$ is a unit vector, called the normalization of u
- 7. $\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2$
 $\|u - v\|^2 = \|u\|^2 - 2\langle u, v \rangle + \|v\|^2$

Theorem 9.1 (Cauchy-Schwarz Inequality). Let V be an inner product space and $u, v \in V$. Then $|\langle u, v \rangle| \leq \|u\|\|v\|$.

Remark 9.3. Let u and v be nonzero vectors in an inner product space V

- 1.
$$-\|u\|\|v\| \leq \langle u, v \rangle \leq \|u\|\|v\|$$

$$-1 \leq \langle u, v \rangle \leq 1$$

- 2. The angle θ between u and v is defined as

$$\theta = \cos^{-1} \left(\frac{\langle u, v \rangle}{\|u\|\|v\|} \right)$$

⁸Also known as a Euclidean space.

Theorem 9.2 (Properties of the Norm). Let V be an inner product space, $u, v \in V$, and $c \in \mathbb{R}$.

1. $\|u\| \geq 0$
2. $\|u\| = 0$ iff $u = \vec{0}$
3. $\|cu\| = |c|\|u\|$
4. $\|u + v\| \leq \|u\| + \|v\|$ (Triangle inequality)

Proof. Recall that, $\|u + v\|^2$ can be expressed as $\langle u + v, u + v \rangle$,

$$\langle u + v, u + v \rangle = \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 \leq (\|u\| + \|v\|)^2$$

■

Remark 9.4 (Properties of Distance). Let V be an inner product space and $u, v, w \in V$. Then

1. $d(u, v) \geq 0$
2. $d(u, v) = 0$ iff $u = v$
3. $d(u, v) = d(v, u)$
4. $d(u, w) \leq d(u, v) + d(v, w)$

Definition 9.3. Let V be an inner product space, $u, v \in V$ and $S \subseteq V$ be nonempty.

1. u is **orthogonal** to v , denoted by $u \perp v$, iff $\langle u, v \rangle = 0$
2. S is an **orthogonal set** if for all $u, v \in V$, $u \neq v$, and $u \perp v$
3. S is an **orthonormal set** if S is orthogonal and every vector in S is a unit vector.

Example 9.1. Let $V = \mathbb{R}^3$ under the standard inner product and

$$S = \{(0, 1, 0), (1, 0, 1), (1, 0, -1)\}, \quad T = \left\{ (0, 1, 0), \left(-\frac{4}{5}, 0, \frac{3}{5}\right), \left(\frac{3}{5}, 0, \frac{4}{5}\right) \right\}$$

If we take all 2-combination of S , then we can show that every pair is orthogonal to each other, but not orthonormal. However, in T , we can show that every pair is orthogonal to each other and each vector is a unit vector, thus, it is orthonormal.

Example 9.2. Let $V = C[-\pi, \pi]$, then $S = \{\sin x, \cos x\}$ is orthogonal, but not orthonormal.⁹

Remark 9.5. Let V be an inner product space:

1. For all $u \in V$, $\vec{0} \perp u$
2. If $u \in V$ such that $u \perp s$, for all $s \in V$, then $u = \vec{0}$
3. If $u \perp v$ and $w \perp v$, then $au + bw \perp v$, for all $a, b \in \mathbb{R}$.
4. $S = \{v_1, v_2, \dots, v_n\}$ is an orthogonal set iff

$$\langle v_i, v_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ \|v_i\|^2 & \text{if } i = j \end{cases} = \delta_{ij}$$

Theorem 9.3 (Pythagorean Theorem). Let V be an inner product space and $u, v \in V$. Then $u \perp v$ iff $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

⁹Prove this as an exercise

Definition 9.4. Let V be an inner product space, $u \in V$, and $S \subseteq V$

1. u is orthogonal to S , denoted by $u \perp S$, if $u \perp s$ for all $s \in S$
2. The orthogonal complement of S , denoted by $S^\perp$ (S-perp) is

$$S^\perp = \{u \in V \mid u \perp s, \forall s \in S\}$$

Remark 9.6. Let V be an inner product space

1. (a) $\{0\}^\perp = \{u \in V \mid \vec{0} \perp u\} = V$
 (b) $V^\perp = \{u \in V \mid u \perp s, \forall s \in S\} = \{\vec{0}\}$
2. If $S \subseteq V$, then $S^\perp = (\text{span } S)^\perp$. In particular if $W = \text{span } S$, then $W^\perp = S^\perp$

Theorem 9.4. If S is a subset of an inner product space, then $S^\perp \leq V$.

Example 9.3. Find the orthogonal complement of $W_1 = \{(c - 6d, d, c) \mid c, d \in \mathbb{R}\} \subseteq \mathbb{R}^3$ under the standard inner product.

Solution.

$$W_1 = \{c(1, 0, 1) + d(-6, 1, 0) \mid c, d \in \mathbb{R}\} = \text{span}\{(1, 0, 1), (-6, 1, 0)\}$$

Thus, $W_1^\perp = S^\perp$ where $S = \{(1, 0, 1), (-6, 1, 0)\}$. Now recall that for a vector to be in the orthogonal complement,

$$(x, y, z) \in S^\perp \iff \begin{cases} \langle (x, y, z), (1, 0, 1) \rangle = 0 \\ \langle (x, y, z), (-6, 1, 0) \rangle = 0 \end{cases}$$

Forming the system of equations,

$$\begin{cases} x + z = 0 \\ -6x + y = 0 \end{cases} \iff \begin{bmatrix} 1 & 0 & 0 \\ -6 & 1 & 0 \end{bmatrix} \xrightarrow{EROs} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$$

Let $z = t$, then $x = -t$ and $y = -6t$,

$$\begin{aligned} W_1^\perp = S^\perp &= \{(-t, -6t, t) \mid t \in \mathbb{R}\} = \{t(-1, -6, 1) \mid t \in \mathbb{R}\} \\ &= \boxed{\text{span}\{(-1, 6, 1)\}} \end{aligned}$$

Theorem 9.5. A finite orthogonal set $S = \{v_1, v_2, \dots, v_n\}$ of nonzero vectors in an inner product space V is linearly independent.

Proof. Suppose that,

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = \vec{0}$$

To demonstrate that $S = \{v_1, v_2, \dots, v_n\}$ is linearly independent, we must show that $c_1 = c_2 = \dots = c_n = 0$. Recall that,

$$\begin{aligned} \langle \vec{0}, v_i \rangle &= 0 \implies \langle c_1 v_1 + c_2 v_2 + \dots + c_n v_n, v_i \rangle \\ &= c_1 \langle v_1, v_i \rangle + c_2 \langle v_2, v_i \rangle + \dots + c_n \langle v_n, v_i \rangle \end{aligned}$$

From the orthogonality of S it follows that $\langle v_i, v_j \rangle = 0$ if $i \neq j$, and it is nonzero if $i = j$. Thus, the equation reduces to

$$c_i \langle v_i, v_i \rangle = 0$$

Since S does not contain nonzero vectors, then this implies that each c_i in the equation is equal to zero. ■

Definition 9.5. Let V be a finite-dimensional inner product space. A basis B for V is called an **orthogonal basis** (resp. orthonormal basis) for V if B is an orthogonal (resp. orthonormal) set.

Theorem 9.6. Let V be an inner product space and $B = \{v_1, v_2, \dots, v_n\}$ be an orthogonal basis for V . Then every $v \in V$ can be written as a linear combination

$$v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

where

$$c_i = \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle}$$

Example 9.4. Let $B = \{[-1 \ 1 \ 1 \ 1]^\top, [1 \ -1 \ 1 \ 1]^\top, [1 \ 1 \ -1 \ 1]^\top, [1 \ 1 \ 1 \ -1]^\top\}$.

1. Show that B is an orthogonal basis of $M_{4 \times 1}(\mathbb{R})$ under the standard inner product.

Solution.

$$\begin{aligned} \langle [-1 \ 1 \ 1 \ 1]^\top, [1 \ -1 \ 1 \ 1]^\top \rangle &= 0, & \langle [-1 \ 1 \ 1 \ 1]^\top, [1 \ 1 \ -1 \ 1]^\top \rangle &= 0 \\ \langle [-1 \ 1 \ 1 \ 1]^\top, [1 \ 1 \ 1 \ -1]^\top \rangle &= 0, & \langle [1 \ -1 \ 1 \ 1]^\top, [1 \ 1 \ -1 \ 1]^\top \rangle &= 0 \\ \langle [1 \ -1 \ 1 \ 1]^\top, [1 \ 1 \ 1 \ -1]^\top \rangle &= 0, & \langle [1 \ 1 \ -1 \ 1]^\top, [1 \ 1 \ 1 \ -1]^\top \rangle &= 0 \end{aligned}$$

2. Use (1) to solve for c in

$$\begin{cases} -a + b + c + d = 1 \\ a - b + c + d = 8 \\ a + b - c + d = -7 \\ a + b + c - d = 2 \end{cases}$$

Solution. The given system is equivalent to

$$a \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \\ 1 \\ 1 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ -7 \\ 2 \end{bmatrix}$$

The vector $[1 \ 8 \ -7 \ 2]^\top$ is a linear combination of the orthogonal set B . By the preceding theorem,

$$c = \frac{\langle [1 \ 8 \ -7 \ 2]^\top, [1 \ 1 \ -1 \ 1]^\top \rangle}{\langle [1 \ 1 \ -1 \ 1]^\top, [1 \ 1 \ -1 \ 1]^\top \rangle} = \frac{18}{4} = \boxed{\frac{9}{2}}$$

Remark 9.7. If $A \in M_n(\mathbb{R})$ has orthogonal columns $a_1, a_2, \dots, a_n$, then the matrix equation $Ax = b$ has a unique solution $x = [x_1 \ x_2 \ \dots \ x_n]^\top$ given by

$$x_i = \frac{\langle b, a_i \rangle}{\langle a_i, a_i \rangle}$$

Theorem 9.7. If $A \in M_n(\mathbb{R})$ has orthogonal columns, then A is nonsingular (If A is orthonormal, then A is nonsingular and its inverse $A^{-1} = A^\top$).

Definition 9.6. Let $u, v \in V$ with $v \neq \vec{0}$ and $u \neq \text{span}\{v\}$. The vector

$$\text{proj}_v u = \frac{\langle u, v \rangle}{\langle v, v \rangle} v$$

is called the (orthogonal) projection of u onto v .

Remark 9.8.

1. $\text{proj}_v u \in \text{span}\{v\}$
2. $u - \text{proj}_v u \perp v$

Definition 9.7. Let $S = \{w_1, w_2, \dots, w_n\}$ be an orthogonal subset of V such that $\vec{0} \notin S$ and let $u \notin \text{span } S$. The vector

$$\text{proj}_S u = \sum_{i=1}^n \text{proj}_{w_i} u = \sum_{i=1}^n \frac{\langle u, w_i \rangle}{\langle w_i, w_i \rangle} w_i$$

is called the **orthogonal projection** of u onto S .

Remark 9.9.

1. $\text{proj}_S u \in \text{span } S$
2. $u - \text{proj}_S u \perp S$

10 Gram-Schmidt Orthonormalization Process

Definition 10.1 (Gram-Schmidt Orthonormalization). Let $B = \{v_1, v_2, \dots, v_n\}$ be a basis for V .

1. Put $w_1 = v_1$.
2. Put $w_2 = v_2 - \text{proj}_{w_1} v_2$.
3. Put $w_3 = v_3 - \text{proj}_{\{w_1, w_2\}} v_3$
4. Combine the above process to get an orthogonal set

$$\{w_1, w_2, \dots, w_n\} = S$$

5. Put $w_n = v_n - \text{proj}_S v_n$,

$$B' = \{w_1, w_2, \dots, w_n\}$$

6. Put $b_i = \frac{w_i}{\|w_i\|}$. Then $B'' = \{b_1, b_2, \dots, b_n\}$ is an orthonormal basis for V .

Remark 10.1.

1. The output of a Gram-Schmidt Process is not unique. Hence a permutation π could lead to a different basis.
2. The recursive definition of the process is:

$$w_k = v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i \rangle}{\langle w_i, w_i \rangle} w_i$$

Theorem 10.1. Every finite-dimensional inner product space V has an orthonormal basis.

Example 10.1. Transform the basis $S = \{(1, 1, 1), (1, -1, -1), (0, 1, 0)\}$ for $\mathbb{R}^3$ into an orthonormal basis under the standard inner product.

Solution. Let us apply Gram-Schmidt Process,

- Put $w_1 = v_1 = (1, 1, 1)$
- Put $w_2 = v_2 - \text{proj}_{w_1} v_2$

$$\begin{aligned} &= v_2 - \frac{\langle w_1, v_2 \rangle}{\langle w_1, w_1 \rangle} w_1 \\ &= (1, -1, -1) + \frac{1}{3}(1, 1, 1) = \left(\frac{4}{3}, -\frac{2}{3}, -\frac{2}{3}\right) \end{aligned}$$

- Put $w_3 = v_3 - \text{proj}_{w_1} v_3 - \text{proj}_{w_2} v_3$

$$\begin{aligned} &= v_3 - \frac{\langle w_1, v_3 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle w_2, v_3 \rangle}{\langle w_2, w_2 \rangle} w_2 \\ &= (0, 1, 0) - \frac{1}{3}(1, 1, 1) + \frac{1}{4} \left(\frac{4}{3}, -\frac{2}{3}, -\frac{2}{3}\right) \\ &= \left(0, \frac{1}{2}, -\frac{1}{2}\right) \end{aligned}$$

Normalizing each w_i ,

- $w_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$

- $w_2 = \sqrt{\frac{3}{8}} \left(\frac{4}{3}, -\frac{2}{3}, -\frac{2}{3} \right)$

- $w_3 = \sqrt{2} \left(0, \frac{1}{2}, -\frac{1}{2} \right)$

Therefore, $B = \left\{ \frac{1}{\sqrt{3}}(1, 1, 1), \sqrt{\frac{3}{8}} \left(\frac{4}{3}, -\frac{2}{3}, -\frac{2}{3} \right), \sqrt{2} \left(0, \frac{1}{2}, -\frac{1}{2} \right) \right\}$ is an orthonormal basis for $\mathbb{R}^3$.

11 Linear Transformations

Definition 11.1. Let U and V , be vector spaces. A **linear transformation** from U to V is a function $T : U \rightarrow V$ satisfying the following:

1. $\forall u_1, u_2 \in U, \quad T(u_1 + u_2) = T(u_1) + T(u_2)$
2. $\forall u \in U, \forall c \in \mathbb{R}, \quad T(cu) = c \cdot T(u)$

Remark 11.1.

1. If $U = V$, then T is called a **linear operator** on V .
2. Let $T : U \rightarrow V$ be a linear transformation

$$(a) \quad \forall u_1, u_2 \in U, \quad T(u_1 - u_2) = T(u_1) - T(u_2)$$

$$(b) \quad \text{Denote by } \vec{0}_u \text{ and } \vec{0}_v \text{ the respective zero vectors of } U \text{ and } V. \text{ Then } T(\vec{0}_u) = \vec{0}_v.$$

$$\text{Therefore, } T(\vec{0}_u + \vec{0}_u) = \vec{0}_v = T(\vec{0}_u)$$

3. $T : U \rightarrow V$ is a linear transformation iff $\forall u_1, u_2 \in U, \forall a, b \in \mathbb{R}$,

$$T(au_1 + bu_2) = aT(u_1) + bT(u_2)$$

Example 11.1. Let U and V be vector spaces. The following are linear transformations:

1. $T : U \rightarrow V$ given by $T(u) = \vec{0}_v, \forall u \in U$.

Solution. Let $u_1, u_2 \in U$ and $c \in \mathbb{R}$. Then,

$$T(u_1 + u_2) = \vec{0}_v$$

$$T(u_1) + T(u_2) = \vec{0}_v + \vec{0}_v = \vec{0}_v = T(u_1 + u_2)$$

$$T(cu_1) = \vec{0}_v$$

$$c \cdot T(u_1) = c\vec{0}_v = \vec{0}_v = T(cu_1)$$

Therefore, T is a linear transformation.

2. $\text{id}_u : U \rightarrow U$ given by $\text{id}_u(u) = u$

Solution. Let $u_1, u_2 \in U$ and $a, b \in \mathbb{R}$. Then,

$$\text{id}_u(au_1 + bu_2) = au_1 + bu_2$$

$$= a \text{id}_u(u_1) + b \text{id}_u(u_2)$$

Therefore, id_u is a linear transformation.

Example 11.2. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x, y)$.

Solution. Let $u_1 = (x_1, y_1, z_1), u_2 = (x_2, y_2, z_2) \in \mathbb{R}^3$ and $a, b \in \mathbb{R}$. Then,

$$T(au_1 + bu_2) = T(ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2)$$

$$= (ax_1 + bx_2, ay_1 + by_2)$$

$$= a(x_1, y_1) + b(x_2, y_2)$$

$$= aT(x_1, y_1, z_1) + bT(x_2, y_2, z_2)$$

$$= aT(u_1) + bT(u_2)$$

Therefore, T is a linear transformation.

Example 11.3. Let $A \in M_{m \times n}(\mathbb{R})$ and $T_A : M_{n \times 1}(\mathbb{R}) \rightarrow M_{m \times 1}(\mathbb{R})$ be defined by $T_A(x) = Ax$.

Solution. Let $x, y \in M_{n \times 1}(\mathbb{R})$ and $a, b \in \mathbb{R}$. Then,

$$\begin{aligned} T_A(ax + by) &= A(ax + by) \\ &= aAx + bAy \\ &= aT_A(x) + bT_A(y) \end{aligned}$$

Therefore, T_A is a linear transformation.

Example 11.4. Let $n \in \mathbb{Z}^+$ and $D : P_n \rightarrow P_{n-1}$, be defined by $D(p(x)) = p'(x)$

Solution. Let $p(x), q(x) \in P_n$ and $c, d \in \mathbb{R}$. Then

$$\begin{aligned} D(cp(x) + dq(x)) &= \frac{d}{dx}(cp(x) + dq(x)) \\ &= cp'(x) + dq'(x) \\ &= cD(p(x)) + dD(q(x)) \end{aligned}$$

Therefore, D is a linear transformation.

Now, let us consider some examples which uses counterexample to prove that a function is **NOT** a linear transformation.

Example 11.5. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = |x|$.

Solution. Take $u_1 = 2, u_2 = -2$. Then,

$$\begin{aligned} T(u_1 + u_2) &= T(0) = |0| = 0 \\ T(u_1) + T(u_2) &= |-2| + |2| = 4 \\ \therefore T(u_1 + u_2) &\neq T(u_1) + T(u_2) \end{aligned}$$

Therefore, T is not a linear transformation.

Example 11.6. Let $T : \mathbb{R}^2 \rightarrow P_2$ given by $T(a, b) = x^2 + ax + b$.

Solution. Let $(a, b) = (6, 7)$ and $c = 2$. Then,

$$\begin{aligned} T(c(6, 7)) &= T(6c, 7c) = T(12, 14) = x^2 + 12x + 14 \\ cT(6, 7) &= 2(x^2 + 6x + 7) = 2x^2 + 12x + 14 \\ \therefore T(cu_1) &\neq cT(u_1) \end{aligned}$$

Therefore, T is not a linear transformation.

Now suppose we flip it around, and we would like to know $T(u)$ where T is a linear transformation, U is a vector space, and $u \in U$. How can we derive this? It turns out that we can do so by rewriting a vector $u \in U$ as a **linear combination**! Recall that $u = c_1u_1 + c_2u_2 + \cdots + c_nu_n$ for some $c_1, c_2, \dots, c_n \in \mathbb{R}$.

$$\begin{aligned} T(u) &= T(c_1u_1 + c_2u_2 + \cdots + c_nu_n) \\ &= c_1T(u_1) + c_2T(u_2) + \cdots + c_nT(u_n) \end{aligned}$$

Therefore, $T(u_1), T(u_2), \dots, T(u_n)$ are sufficient to determine $T(u)$. Then, T is said to be **completely determined** by $T(u_1), T(u_2), \dots, T(u_n)$.

Example 11.7. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(1, 0) = (2, 1)$ and $T(0, 1) = (-3, 2)$. Find $T(x, y)$ for any $x, y \in \mathbb{R}$.

Solution. Let $(x, y) \in \mathbb{R}^2$. Then,

$$\begin{aligned} T(x, y) &= T(x(1, 0) + y(0, 1)) \\ &= xT(1, 0) + yT(0, 1) \\ &= x(2, 1) + y(-3, 2) \\ &= \boxed{(2x - 3y, x + 2y)} \end{aligned}$$

Definition 11.2. Let $T : U \rightarrow V$ be a linear transformation.

1. The **kernel** of T is $\ker T = \{u \in U \mid T(u) = \vec{0}_v\} \subseteq U$.
2. The **image/range** of T is $\text{im } T = \{T(u) \mid u \in U\} \subseteq V$.

Remark 11.2. Let $u \in U$ and $v \in V$.

1. $u \in \ker T$ iff $T(u) = \vec{0}_v$
2. $v \in \text{im } T$ iff $v = T(u)$ for some $u \in U$

Theorem 11.1. Let $T : U \rightarrow V$ be a linear transformation. Then $\ker T \leq U$ and $\text{im } T \leq V$.

Definition 11.3. Let $T : U \rightarrow V$ be a linear transformation with U and V be finite-dimensional.

1. The **rank** of T is $\text{rank } T = \dim(\text{im } T)$
2. The **nullity** of T is $v(T) = \dim(\ker T)$

Example 11.8. Let $A \in M_{m \times n}(\mathbb{R})$ and $T_A : M_{n \times 1}(\mathbb{R}) \rightarrow M_{m \times 1}(\mathbb{R})$ defined by $T_A(x) = Ax$.

(a)

$$\begin{aligned} \ker T_A &= \{x \in M_{n \times 1}(\mathbb{R}) \mid T_A(x) = 0_{m \times 1}\} \\ &= \{x \in M_{n \times 1}(\mathbb{R}) \mid Ax = 0_{m \times 1}\} \\ &= N(A), \quad \text{the null space of } A \\ \therefore v(T_A) &= \dim(\ker T_A) = \dim N(A) = v(A) \end{aligned}$$

(b)

$$\begin{aligned} \text{im } T_A &= \{T_A(x) \mid x \in M_{n \times 1}(\mathbb{R})\} \\ &= \{Ax \mid x \in M_{n \times 1}(\mathbb{R})\} \\ &= \{x_1 C_1 + x_2 C_2 + \cdots + x_n C_n \mid x_1, x_2, \dots, x_n \in \mathbb{R} \text{ and } C_i = \text{col } i \text{ of } A\} \\ &= \text{span}\{C_1, C_2, \dots, C_n\} \\ &= C(A), \quad \text{the column space of } A \\ \therefore \text{rank}(T_A) &= \dim(\text{im } T) = \dim C(A) = \text{rank } A \end{aligned}$$

Example 11.9. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + z, x + y, y - z)$.¹⁰ Find the bases for the image and kernel of T .

Solution.

(a) Recall that $(x, y, z) \in \ker T$ iff $T(x, y, z) = (0, 0, 0)$. Then,

$$(x + z, x + y, y - z) = (0, 0, 0) \iff \begin{cases} x + z = 0 \\ x + y = 0 \\ y - z = 0 \end{cases}$$

Turning this into a coefficient matrix,

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \iff \begin{cases} x + z = 0 \\ y - z = 0 \end{cases}$$

Since the third column is a free variable, we let $z = t$. Then, $x = -t$ and $y = t$. Therefore $\ker T = \{(-t, t, t) \mid t \in \mathbb{R}\} = \text{span}\{(-1, 1, 1)\}$.

Then $B = \{(-1, 1, 1)\}$ is a basis for $\ker T$, and $v(T) = 1$.

(b) Let us find a basis for $\text{im } T$,

$$\begin{aligned} \text{im } T &= \{T(x, y, z) \mid x, y, z \in \mathbb{R}\} \\ &= \{(x + z, x + y, y - z) \mid x, y, z \in \mathbb{R}\} \\ &= \{x(1, 1, 0) + y(0, 1, 1) + z(1, 0, -1) \mid x, y, z \in \mathbb{R}\} \\ &= \text{span}\{(1, 1, 0), (0, 1, 1), (1, 0, -1)\} \end{aligned}$$

Then, let us check whether this is linearly independent. Form,

$$x(1, 1, 0) + y(0, 1, 1) + z(1, 0, -1) = (0, 0, 0)$$

The coefficient matrix is,

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Since there is a free variable in the third column, we must delete that.

Therefore $B = \{(1, 1, 0), (0, 1, 1)\}$ is a basis for $\text{im } T$ and $\text{rank } T = 2$

Theorem 11.2 (Rank-Nullity Theorem). Let U and V be vector spaces with $\dim U = n < \infty$. If $T : U \rightarrow V$ is a linear transformation, then

$$\dim U = \text{rank } T + v(T)$$

Definition 11.4. Let $T : U \rightarrow V$ be a linear transformation.

1. T is **one-to-one** if $\forall u_1, u_2 \in U$. $T(u_1) = T(u_2)$ iff $u_1 = u_2$.
2. T is **onto** if $\text{im } T = V$, i.e. $\forall v \in V$, $\exists u \in U$ such that $v = T(u)$.
3. T is a **bijection/isomorphism** if T is both one-to-one and onto.

¹⁰As an exercise prove that this is a linear transformation.

Remark 11.3. Let U and V be vector spaces,

1. If there exists an isomorphism $T : U \rightarrow V$, then U and V are said to be isomorphic, denoted by $U \cong V$.
2. If $T : U \rightarrow V$ is a linear transformation, then T is one-to-one iff $\ker T = \{\vec{0}_v\}$. Only contains the zero vector.
3. T is onto iff $\text{rank } T = \dim V$.

For the next few examples, we will **determine the mapping** of a given function using the remarks given above.

Example 11.10. Determine the mapping of a function $\text{id}_u : U \rightarrow U$ defined by $\text{id}_u(u) = u$.

Solution.

1. $u \in \ker(\text{id}_u)$ iff $\text{id}_u(u) = \vec{0}_u$ and $u = \vec{0}_u$.

Therefore, $\ker(\text{id}_u) = \vec{0}_u$, hence, id_u is one-to-one.

2. $\text{im}(\text{id}_u) = \{\text{id}_u(u) \mid u \in U\} = \{u \mid u \in U\}$

Therefore id_u is onto. And since id_u is both one-to-one and onto, it is an isomorphism.

Example 11.11. Determine the mapping of a function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x, y, 0)$.

Solution.

- (a) $(x, y) \in \ker T$ iff $T(x, y) = (0, 0, 0)$

$$(x, y, 0) = (0, 0, 0) \implies x = y = 0$$

Therefore, $\ker T = \{\vec{0}\}$, hence, T is one-to-one.

- (b) $\text{im } T = \{T(x, y) \mid x, y \in \mathbb{R}\}$

$$\begin{aligned} &= \{(x, y, 0) \mid x, y \in \mathbb{R}\} \\ &= \{x(1, 0, 0) + y(0, 1, 0) \mid x, y \in \mathbb{R}\} \\ &= \text{span}\{(1, 0, 0), (0, 1, 0)\} \end{aligned}$$

Since neither of them are scalar multiples of each other it is a basis for $\text{im } T$. And since $\text{rank } T \neq \dim \mathbb{R}^3$, it is not onto.

Therefore, T is one-to-one.

Example 11.12. Determine the mapping of a function $T : \mathbb{R}^3 \rightarrow P_2$ defined by $T(a, b, c) = (a + b)x^2 + (b + c)x + c$.

Solution.

- (a) $(a, b, c) \in \ker T$ iff $(a + b)x^2 + (b + c)x + c = 0$

$$\begin{cases} a + b = 0 \\ b + c = 0 \\ c = 0 \end{cases} \iff (a, b, c) = (0, 0, 0)$$

Therefore $\ker T = \{(0, 0, 0)\}$, hence, T is one-to-one.

$$\begin{aligned}
(b) \quad \text{im } T &= \{T(a, b, c) \mid a, b, c \in \mathbb{R}\} \\
&= \{(a+b)x^2 + (b+c)x + c \mid a, b, c \in \mathbb{R}\} \\
&= \{a(x^2) + b(x^2 + x) + c(x+1) \mid a, b, c \in \mathbb{R}\} \\
&= \text{span}\underbrace{\{x^2, x^2 + x, x + 1\}}_B
\end{aligned}$$

We can verify that B is a basis for $\text{im } T$. And since $\text{rank } T = 3 = \dim P_2$, hence, T is onto.

Therefore, T is an isomorphism, and $\mathbb{R}^3 \cong P_2$.

Theorem 11.3. Let U, V, W be vector spaces and $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transformations.

1. The function $S \circ T : U \rightarrow W$ is a linear transformation. If S and T are isomorphic, then so is $S \circ T$.
2. T is an isomorphism iff there exists a unique function $T^{-1} : V \rightarrow U$ such that,

$$(T \circ T^{-1})(v) = v, \quad \forall v \in V$$

$$(T^{-1} \circ T)(u) = u, \quad \forall u \in U$$

Remark 11.4. Let U, V, W be vector spaces

1. If $T : U \rightarrow V$ is an isomorphism, then the function $T^{-1} : V \rightarrow U$ is also an isomorphism, called the inverse of T .
2. (a) $U \cong U$
(b) If $U \cong V$, then $V \cong U$
(c) If $U \cong V$ and $V \cong W$, then $U \cong W$

Theorem 11.4. Let U and V be finite-dimensional vector spaces.

1. If $\dim V = n$, then $V \cong \mathbb{R}^n$
2. $U \cong V$ iff $\dim U = \dim V$
3. Let $T : U \rightarrow V$ be a linear transformation. If $\dim U = \dim V$, then the following are equivalent:
 - (a) T is one-to-one
 - (b) T is onto
 - (c) T is an isomorphism

Example 11.13.

1. $M_{n \times 1}(\mathbb{R}) \cong \mathbb{R}^n$
2. $M_{m \times n}(\mathbb{R}) \cong M_{n \times m}(\mathbb{R}) \cong \mathbb{R}^{mn}$
3. $P_n \cong \mathbb{R}^{n+1}$
4. Show that $T : \mathbb{R}^2 \rightarrow \mathbb{C}$ given by $T(a, b) = (a + b) + (a - b)i$ is an isomorphism.

Remark 11.5. Let $T : U \rightarrow V$ be a linear transformation.

1. T is not necessarily an isomorphism even when $U \cong V$. It only guarantees that such functions exist.
2. If $\dim U \neq \dim V$, then T is **NOT** an isomorphism.
 - (a) If $\dim U > \dim V$, then T is not one-to-one.
 - (b) If $\dim U < \dim V$, then T is not onto.

12 Matrix of Linear Transformations

Definition 12.1. Let V be a finite-dimensional vector space. An **ordered basis** for V such that the order in which the elements are listed is taken into account.

Example 12.1. In $\mathbb{R}^3$, $B_1 = \{e_1, e_2, e_3\}$ and $B_2 = \{e_3, e_1, e_2\}$ are equal sets, but not ordered bases.

Example 12.2. Let $V = \mathbb{R}^2$ with ordered bases

$$B_1 = \{(1, 0), (0, 1)\}$$

$$B_2 = \{(0, 1), (1, 0)\}$$

$$B_3 = \{(1, 2), (3, 5)\}$$

1. If $v = (-2, 5)$, find $[v]_{B_i}$, $i \in \{1, 2, 3\}$

Solution.

- $v = (-2, 5) = -2(1, 0) + 5(0, 1)$

$$[v]_{B_1} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

$$[v]_{B_2} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

- To find $[v]_{B_3}$,

$$v = (-2, 5) = x(1, 2) + y(3, 5) = (x + 3y, 2x + 5y)$$

$$\left[\begin{array}{cc|c} 1 & 3 & -2 \\ 2 & 5 & 5 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & 0 & 25 \\ 0 & 1 & -9 \end{array} \right]$$

$$[v]_{B_3} = \begin{bmatrix} 25 \\ -9 \end{bmatrix}$$

2. If $w \in \mathbb{R}^2$ such that $[w]_{B_3} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$, find w .

Solution.

$$w = -1(1, 2) + 3(3, 5) = \boxed{(8, 13)}$$

Definition 12.2. Let U and V be vector spaces with respective ordered bases

$$B_U = \{u_1, u_2, \dots, u_n\}$$

$$B_V = \{v_1, v_2, \dots, v_m\}$$

If $T : U \rightarrow V$ is a linear transformation, the **matrix representation of T with respect to B_U and B_V** is

$$[T]_{B_V}^{B_U} = \begin{bmatrix} [T(u_1)]_{B_V} & [T(u_2)]_{B_V} & \cdots & [T(u_n)]_{B_V} \end{bmatrix} \in M_{m \times n}(\mathbb{R})$$

Example 12.3. Let V be a 3-dimensional vector space with ordered basis $B = \{v_1, v_2, v_3\}$. If $\text{id}_V : V \rightarrow V$ is the identity linear transformation, find $[\text{id}_V]_B^B$.

Solution.

$$\begin{aligned} [\text{id}_V]_B^B &= \begin{bmatrix} [\text{id}_V(v_1)]_B & [\text{id}_V(v_2)]_B & [\text{id}_V(v_3)]_B \end{bmatrix} \\ &= \begin{bmatrix} [v_1]_B & [v_2]_B & [v_3]_B \end{bmatrix} \end{aligned}$$

Then we must find the values in B under id_V ,

$$v_1 = 1v_1 + 0v_2 + 0v_3$$

$$v_2 = 0v_1 + 1v_2 + 0v_3$$

$$v_3 = 0v_1 + 0v_2 + 1v_3$$

$$\therefore [\text{id}_V]_B^B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example 12.4. Let $B_2 = \{(1, 3), (-2, 4)\}$ and $B_3 = \{(1, 1, 1), (2, 2, 0), (3, 0, 0)\}$ be the ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is the linear transformation given by

$$T(x, y) = (x + 2y, -x, 0)$$

Find $[T]_{B_2}^{B_3}$.

Remark 12.1. Let U and V be finite-dimensional vector spaces with respective ordered bases $B_U = \{u_1, u_2, \dots, u_n\}$ and $B_V = \{v_1, v_2, \dots, v_m\}$ and $T : U \rightarrow V$ be a linear transformation. To find $[T]_{B_U}^{B_V}$:

1. Find $T(u_1), T(u_2), \dots, T(u_n)$
2. Form the augmented matrix

$$X = [B_V \mid T(B_U)] = \begin{bmatrix} v_1 & v_2 & \cdots & v_n & \mid & T(u_1) & T(u_2) & \cdots & T(u_n) \end{bmatrix}$$

3. Use elementary row operations to transform X into the matrix $[I_n \mid M]$. Then $M = [T]_{B_U}^{B_V}$.

Theorem 12.1. Let U and V be finite-dimensional vector spaces with ordered bases B_U and B_V , respectively. If $T : U \rightarrow V$ is a linear transformation and $u \in U$, then

$$[T(u)]_{B_V} = [T]_{B_U}^{B_V} [u]_{B_U}$$

Example 12.5. Consider the ordered bases $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and $C = \{x^2, x, 1\}$ for $\mathbb{R}^3$ and P_2 , respectively. If $T : \mathbb{R}^3 \rightarrow P_2$ is the linear transformation such that

$$[T]_B^C = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

determine $T(a, b, c)$ for all $(a, b, c) \in \mathbb{R}^3$.

Solution. Let $u = (a, b, c) \in \mathbb{R}^3$. Then

$$[T(a, b, c)]_C = [T]_B^C [(a, b, c)]_B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b + c \\ a + c \\ a + b \end{bmatrix}$$

Therefore, $T(a, b, c) = (b + c)x^2 + (a + c)x + (a + b)$ for all $a, b, c \in \mathbb{R}$.

Theorem 12.2. Let U, V , and W be finite-dimensional vector spaces with respective ordered bases B_U, B_V , and B_W . If $T : U \rightarrow V$ and $S : V \rightarrow W$ are linear transformations, then

1. $[S \circ T]_{B_U}^{B_W} = [S]_{B_V}^{B_W} [T]_{B_U}^{B_V}$

13 Eigenvalues and Eigenvectors

Definition 13.1. Let $A \in M_n(\mathbb{R})$. Then $\lambda \in \mathbb{R}$ is called an **eigenvalue** of A if there exists nonzero $v \in M_{n \times 1}(\mathbb{R})$ such that

$$Av = \lambda v$$

Such a vector v is called an **eigenvector** of A associated to A .

Remark 13.1. Let $A \in M_n(\mathbb{R})$

1. We impose the condition that $v \notin 0_{n \times 1}$ since $A0_{n \times 1} = \lambda 0_{n \times 1}$ is true for any $\lambda \in \mathbb{R}$.
2. v is an eigenvector of A iff v is a scalar multiple of Av .

Example 13.1.

1. Let $v \in M_{n \times 1}(\mathbb{R})$ be nonzero

- $I_n v = 1v$

Therefore, $\lambda = 1$ is an eigenvalue of I_n where associated eigenvectors are any nonzero $v \in M_{n \times 1}(\mathbb{R})$

- $0_{n \times n} v = 0_{n \times 1} = 0v$

Therefore, $\lambda = 0$ is an eigenvalue of $0_{n \times n}$ where associated eigenvectors are nonzero $v \in M_{n \times 1}(\mathbb{R})$.

2. Let $A = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$

- $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\therefore \lambda_1 = 2$ is an eigenvalue of A with associated eigenvector $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- $A \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\therefore \lambda_2 = -2$ is an eigenvalue of A with associated eigenvector $v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

3. Let $A = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$.

- $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$\therefore \lambda_1 = 4$ is an eigenvalue of A with associated eigenvector $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

- $A \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$\therefore \lambda_2 = -2$ is an eigenvalue of A with associated eigenvector $v_2 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$.

Definition 13.2. Let $A \in M_n(\mathbb{R})$.

1. The **characteristic polynomial** of A is

$$p_A(x) = \det(A - xI_n) = \begin{vmatrix} a_{11} - x & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - x & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - x \end{vmatrix}$$

2. The **characteristic equation** of A is $p_A(x) = 0$.

Remark 13.2. Let $A \in M_n(\mathbb{R})$.

1. $p_A(x)$ is a polynomial of degree n with leading coefficients $(-1)^n$ and constant term $\det A$.
2. $\lambda \in \mathbb{R}$ is an eigenvalue of A iff $p_A(\lambda) = 0$
3. If $p_A(x)$ has no real roots, then A has no (real) eigenvalues.
4. The **algebraic multiplicity** of λ is the multiplicity of λ as a root of $p_A(x)$, denoted by $a(\lambda)$

Definition 13.3. Let $A \in M_n(\mathbb{R})$ and $\lambda \in \mathbb{R}$ be an eigenvalue of A . Then

$$E(A, \lambda) := N(A - \lambda I_n)$$

is called the **eigenspace of A associated to λ** .

Remark 13.3.

1. $E(A, \lambda)$ is an eigenspace of $M_{n \times 1}(\mathbb{R})$. The **geometric multiplicity** of λ is $g(\lambda) = \dim E(A, \lambda) = v(A - \lambda I_n)$.
2. The eigenvectors of A associated to λ are the nonzero vectors in $E(A, \lambda)$. Thus, $\mu \in \mathbb{R}$ is NOT an eigenvalue of A iff $N(A - \mu I_n) = \{\vec{0}\}$.
3. To find (a basis for) $E(A, \lambda)$, solve for v in $(A - \lambda I_n)v = 0_{n \times 1}$ (by GJR/GE).

Example 13.2. Find the eigenvalues and corresponding eigenspaces of a matrix A where,

$$A = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

and identify the algebraic and geometric multiplicity of each eigenvalue.

Solution.

Let us first solve for its characteristic polynomial $p_A(x)$,

$$p_A(x) = \begin{vmatrix} -x & 2 \\ 2 & -x \end{vmatrix} = x^2 - 4$$

Therefore, the eigenvalues of A : $2, -2$

- $E(A, 2)$

Forming the augmented matrix,

$$\left[\begin{array}{cc|c} -2 & 2 & 0 \\ 2 & -2 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right] \longleftrightarrow a - b = 0$$

Since there are no leading entries in the second column, let $b = t$, then, $a = t$,

$$E(A, 2) = \left\{ \begin{bmatrix} t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

Since the set only contains one element, it is a basis for $E(A, 2)$. And $a(2) = 1 = g(2)$.

- $E(A, -2)$

Forming the augmented matrix,

$$\left[\begin{array}{cc|c} 2 & 2 & 0 \\ 2 & 2 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] \longleftrightarrow a + b = 0$$

Let $b = t$, then, $a = -t$,

$$E(A, -2) = \left\{ \begin{bmatrix} -t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

Since the set only contains one element, it is a basis for $E(A, -2)$ and $a(-2) = 1 = g(-2)$.

Example 13.3. Find the eigenvalues and corresponding eigenspaces of a matrix B where,

$$B = \begin{bmatrix} 4 & 2 & 3 \\ 2 & 1 & 2 \\ -1 & -2 & 0 \end{bmatrix}$$

and identify the algebraic and geometric multiplicity of each eigenvalue.

Solution.

Let us first solve for its characteristic polynomial $p_B(x)$,

$$p_B(x) = \begin{vmatrix} 4-x & 2 & 3 \\ 2 & 1-x & 2 \\ -1 & -2 & -x \end{vmatrix} \xrightarrow{\substack{R_1 \leftarrow R_3 + R_1 \\ C_3 \leftarrow C_3 - C_1}} \begin{vmatrix} 3-x & 0 & 0 \\ 2 & 1-x & 0 \\ -1 & -2 & 1-x \end{vmatrix} = (1-x)^2(3-x)$$

Therefore, the eigenvalues of B : 1, 3

- $E(A, 1)$

Forming the augmented matrix,

$$\left[\begin{array}{ccc|c} 3 & 2 & 3 & 0 \\ 2 & 0 & 2 & 0 \\ -1 & -2 & -1 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a + c = 0 \\ b = 0 \end{cases}$$

Let $c = t$, then, $a = -t$,

$$E(A, 1) = \left\{ \begin{bmatrix} -t \\ 0 \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Since it only contains one element it is a basis for the eigenspace. And $a(1) = 2 \neq 1 = g(1)$

- $E(A, 3)$

Forming the augmented matrix,

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & -2 & 2 & 0 \\ -1 & -2 & -3 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a + 2b + 3c = 0 \\ b + \frac{2}{3}c = 0 \end{cases}$$

Let $c = t$, then, $b = -\frac{2}{3}t$, $a = -\frac{5}{3}t$,

$$E(A, 3) = \left\{ \begin{bmatrix} -\frac{5}{3}t \\ -\frac{2}{3}t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} -\frac{5}{3} \\ -\frac{2}{3} \\ 1 \end{bmatrix} \mid t \in \mathbb{R} \right\}$$

Since it only contains one element it is a basis, and $a(3) = 1 = g(3)$.

Example 13.4. Find the eigenvalues and corresponding eigenspaces of a matrix C where,

$$C = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 5 & 1 \end{bmatrix}$$

and identify the algebraic and geometric multiplicity of each eigenvalue.

Let us first determine the characteristic polynomial $p_C(x)$,

$$p_C(x) = \begin{vmatrix} 1-x & 2 & 0 & 0 \\ 0 & 1-x & 0 & 0 \\ 0 & 0 & -1-x & 0 \\ 0 & 0 & 5 & 1-x \end{vmatrix}$$

Performing cofactor expansion along column 1,

$$= (1-x) \begin{vmatrix} 1-x & 0 & 0 \\ 0 & -1-x & 0 \\ 0 & 5 & 1-x \end{vmatrix} = (1-x)^3(-1-x)$$

Therefore, the eigenvalues of C : $1, -1$

- $E(A, 1)$

$$\left[\begin{array}{cccc|c} 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} b = 0 \\ c = 0 \end{cases}$$

Let $a = s$ and $d = t$, then,

$$\begin{aligned} E(A, 1) &= \left\{ \begin{bmatrix} s \\ 0 \\ 0 \\ t \end{bmatrix} \mid s, t \in \mathbb{R} \right\} = \left\{ s \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \mid s, t \in \mathbb{R} \right\} \\ &= \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \end{aligned}$$

Since they are not scalar multiples of each other, it is a basis for the eigenspace. And $a(1) = 3 \neq 2 = g(1)$.

- $E(A, -1)$

$$\left[\begin{array}{cccc|c} 2 & 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 2 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{2}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a + b = 0 \\ b = 0 \\ c + \frac{2}{5}d = 0 \end{cases}$$

Let $d = t$, then, $a = 0, c = -\frac{2}{5}t$,

$$E(A, -1) = \left\{ \begin{bmatrix} 0 \\ 0 \\ -\frac{2}{5}t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ -\frac{2}{5} \\ 1 \end{bmatrix} \right\}$$

Since it only contains one element, it is a basis for the eigenspace. And $a(-1) = 1 = g(-1)$.

Question.

Now, given a square matrix when is it nonsingular if we only know its eigenvalue? Let us observe. Suppose that there exists a matrix $A \in M_n(\mathbb{R})$. Then, A is nonsingular iff $\det A \neq 0$. This implies that $p_A(0) \neq 0$ and 0 is not a root of the characteristic polynomial, hence 0 is not an eigenvalue of A .

Theorem 13.1. Let $A \in M_n(\mathbb{R})$. Then A is nonsingular iff 0 is not an eigenvalue of A .

14 Similarity, Diagonalization, and Symmetric Matrices

Definition 14.1. Let $A, B \in M_n(\mathbb{R})$. Then A and B are **similar** if there exists $P \in M_n(\mathbb{R})$ such that $B = P^{-1}AP$.

Remark 14.1. Let $A, B \in M_n(\mathbb{R})$ be similar, and V be an n -dimensional vector space.

1. Then A and B represent the same linear operator $T : U \rightarrow V$.
2. Then A and B are equivalent, hence $\text{rank } A = \text{rank } B$.

Now, let's explore some of the properties of similar matrices. Suppose that there exists $A, B \in M_n(\mathbb{R})$ such that A and B are similar. Then, it follows that $B = P^{-1}AP$ for some $P \in M_n(\mathbb{R})$.

Since A and B are square matrices, we could obtain their determinant,

$$\begin{aligned}\det B &= \det(P^{-1}AP) \\ &= \det(P^{-1}) \det A \det P \\ &= \frac{1}{\det P} (\det A) (\det P) \\ &= \det A\end{aligned}$$

Now let's observe their characteristic polynomials,

$$\begin{aligned}p_B(x) &= \det(B - xI_n) = \det(P^{-1}AP - xP^{-1}P) \\ &= \det[(P^{-1}A - xP^{-1}(xI_n))P] \\ &= \det[P^{-1}(A - xI_n)P] \\ &= \det(A - xI_n) \\ &= p_A(x)\end{aligned}$$

Theorem 14.1. Let $A, B \in M_n(\mathbb{R})$ be similar. Then

1. $\text{rank } A = \text{rank } B$ and $v(A) = v(B)$
2. $\det A = \det B$
3. $p_A(x) = p_B(x)$ (hence A and B have the same eigenvalues)
4. A^k is similar to B^k for all $k \in \mathbb{Z}^+$
5. $A^\top$ is similar to $B^\top$
6. If A is nonsingular, then B is nonsingular and A^{-1} is similar to B^{-1}

Definition 14.2. $A \in M_n(\mathbb{R})$ is said to be **diagonalizable** if A is similar to a diagonal matrix, i.e. there exists $P \in M_n(\mathbb{R})$ such that $P^{-1}AP$ is diagonal.

An example of this is suppose we have two matrices $A, P \in M_n(\mathbb{R})$, where,

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Then,

$$P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Therefore, A is diagonalizable.

Question. When is $A \in M_n(\mathbb{R})$ diagonalizable?

A is diagonalizable iff there exists a nonsingular matrix $P \in M_n(\mathbb{R})$ such that $P^{-1}AP = D = \text{diag}(d_1, \dots, d_n)$. Let P_i be column i of P . Then $AP = PD$. Now

Theorem 14.2. $A \in M_n(\mathbb{R})$ is diagonalizable iff A has n linearly independent eigenvectors.

To diagonalize $A \in M_n(\mathbb{R})$:

1. Find the eigenvalues λ_i of A .
2. Find bases for each eigenspace $E(A, \lambda_i)$.
3. Form the union E of the bases found in (2). Then E is linearly independent. If $|E| = n$, then A is diagonalizable. Otherwise A is not diagonalizable.
4. If A is diagonalizable, then $P^{-1}AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ where $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$ are the eigenvalues (counting algebraic multiplicities and column i of P is an eigenvector of A associated to λ_i). We will denote the set of eigenvalues as $\sigma(A)$.

Example 14.1. Determine if the matrix A is diagonalizable. If so, find diagonal D and nonsingular P such that $D = P^{-1}AP$. Where,

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}$$

Solution.

We first compute for the characteristic polynomial $p_A(x)$,

$$p_A(x) = \begin{vmatrix} 1-x & 0 & 0 & 0 \\ 0 & 1-x & 5 & -10 \\ 1 & 0 & 2-x & 0 \\ 1 & 0 & 0 & 3-x \end{vmatrix}$$

By performing cofactor expansion along row 1,

$$= 1-x \begin{vmatrix} 1-x & 5 & -10 \\ 0 & 2-x & 0 \\ 0 & 0 & 3-x \end{vmatrix} = (1-x)^2(2-x)(3-x)$$

Therefore, the eigenvalues of A : 1, 2, 3

- $E(A, 1)$

$$\left[\begin{array}{cccc|c} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & -10 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 2 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a + 2d = 0 \\ c - 2d = 0 \end{cases}$$

Let $b = s$ and $d = t$, then, $a = -2t, c = 2t$,

$$E(A, 1) = \left\{ \begin{bmatrix} -2t \\ s \\ 2t \\ t \end{bmatrix} \mid s, t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix} \right\}$$

The two elements are not scalar multiples of each other. Hence, $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix} \right\}$ is a basis for

$E(A, 1)$. With $a(1) = 2 = g(1)$.

- $E(A, 2)$

$$\left[\begin{array}{cccc|c} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 5 & -10 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a + d = 0 \\ b - 5c = 0 \\ d = 0 \end{cases}$$

Let $c = t$, then, $a = 0, b = 5t$,

$$E(A, 2) = \left\{ \begin{bmatrix} 0 \\ 5t \\ t \\ 0 \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 0 \\ 5 \\ 1 \\ 0 \end{bmatrix} \right\}$$

And since there is only one nonzero vector in the set it is a basis for $E(A, 2)$. And $a(2) = 1 = g(2)$.

- $E(A, 3)$

$$\left[\begin{array}{cccc|c} -2 & 0 & 0 & 0 & 0 \\ 0 & -2 & 5 & -10 & 0 \\ 1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 5 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a = 0 \\ b + 5d = 0 \\ c = 0 \end{cases}$$

Let $d = t$, then, $b = -5t$,

$$E(A, 3) = \left\{ \begin{bmatrix} 0 \\ -5t \\ 0 \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 0 \\ -5 \\ 0 \\ 1 \end{bmatrix} \right\}$$

And since there is only one nonzero element in the set, it is a basis for $E(A, 3)$. And $a(3) = 1 = g(3)$.

From this let us form the set E which contains the elements of the basis of each eigenspace $E(A, \lambda_i)$ for all λ_i .¹¹

$$E = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -5 \\ 0 \\ 1 \end{bmatrix} \right\}$$

And since $|E| = 4$, which is equal to the size of A . Thus, A is diagonalizable with $D = P^{-1}AP$ where,

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}, \quad P = \begin{bmatrix} 0 & -2 & 0 & 0 \\ 1 & 0 & 5 & -5 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

¹¹One could check that set E is linearly independent.

The permutations of columns in P must follow the placing of eigenvalues in D . Any permutation for multiplicities (i.e. $\sigma(A) = \{1, 2, 1, 3\} \implies D = \text{diag}(1, 2, 1, 3)$) are allowed. To further illustrate this let's find another solution for the example above.

$$D = \text{diag}(1, 2, 1, 3) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}, \quad P = \begin{bmatrix} 0 & 0 & -2 & 0 \\ 1 & 5 & 0 & -5 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

One could check that this is true,¹² and any permutation of the diagonal entries as long as the placing of vectors from E must follow the order in D .

Theorem 14.3. Let $A \in M_n(\mathbb{R})$. Then A is diagonalizable iff both the following conditions hold:

1. Every root of $p_A(x)$ is real.
2. For every eigenvalue λ of A , $a(\lambda) = g(\lambda)$.

Remark 14.2. Let $A \in M_n(\mathbb{R})$.

1. For every eigenvalue λ of A_i , $a(\lambda) \geq g(\lambda) \geq 1$.
2. If $p_A(x)$ has exactly n distinct real roots, then A is diagonalizable.

Recall.

1. Standard inner product on $M_{n \times 1}(\mathbb{R})$,

$$\langle x, y \rangle = x^\top y, \quad \forall x, y \in M_{n \times 1}(\mathbb{R})$$

2. $S = \{v_1, v_2, \dots, v_n\}$ is an orthonormal set iff $\langle v_i, v_j \rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases} = \delta_{ij}$

Question. When are the columns of $P \in M_n(\mathbb{R})$ orthonormal under the standard inner product of $M_{n \times 1}(\mathbb{R})$?

Let $P \in M_n(\mathbb{R})$, with columns $P_1, P_2, \dots, P_n$. Then $\{P_1, P_2, \dots, P_n\}$ is an orthonormal set iff:

$$\langle P_i, P_j \rangle = \delta_{ij}, \quad \forall i, j$$

$$P_i^\top P_j = \delta_{ij}, \quad \forall i, j$$

$$(\text{row } i \text{ of } P^\top)(\text{column } j \text{ of } P) = \delta_{ij}, \quad \forall i, j$$

$$P^\top P = I_n$$

Definition 14.3. $P \in M_n(\mathbb{R})$ is **orthogonal** if $P^\top P = I_n$.

Theorem 14.4. Let $P \in M_n(\mathbb{R})$. Then the following are equivalent:

1. P is orthogonal.
2. P is nonsingular with $P^{-1} = P^\top$.
3. The columns of P form an orthonormal basis for $M_{n \times 1}(\mathbb{R})$ (under the standard inner product).

¹²This can be easily verified using Matlab or Numpy.

Example 14.2. Shown below are some examples of orthogonal matrices that can be verified by simply pairing the rows and getting their inner product (respectively pairing the columns and getting their inner product). It is not hard to show that the inner product is 0.

1.

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad \forall \theta \in \mathbb{R}$$

2.

$$\begin{bmatrix} -2/3 & 2/3 & 1/3 \\ 2/3 & 1/3 & 2/3 \\ 1/3 & 2/3 & -2/3 \end{bmatrix}$$

Remark 14.3. Let $P, Q \in M_n(\mathbb{R})$ be orthogonal

1. Then PQ is also orthogonal

$$(PQ)^\top(PQ) = Q^\top P^\top PQ = I_n$$

2. $\det P = \pm 1$

$$\begin{aligned} (\det P^\top)(\det P) &= \det I_n \\ (\det P)^2 &= 1 \\ \therefore \det P &= \pm 1 \end{aligned}$$

3. Let $D \in M_n(\mathbb{R})$ be diagonal and $A = PDP^\top$. Then

$$\begin{aligned} A^\top &= (PDP^\top)^\top \\ &= (P^\top)^\top D^\top P^\top \\ &= PDP^{-1} = A \end{aligned}$$

Definition 14.4. Let $A \in M_n(\mathbb{R})$. Then

1. A is **symmetric** if $A^\top = A$.
2. A is **orthogonally diagonalizable** if there exists an orthogonal $P \in M_n(\mathbb{R})$ such that $P^\top AP$ is diagonal.

Theorem 14.5 (Spectral Decomposition Theorem for Symmetric Matrices). $A \in M_n(\mathbb{R})$ is orthogonally diagonalizable iff A is symmetric.

To orthogonally diagonalize a symmetric matrix $A \in M_n(\mathbb{R})$.

1. Find the eigenvalues λ_i of A .
2. Find **orthonormal basis** of $E(A, \lambda_i)$ using Gram-Schmidt Process.
3. Form the union $E = \{v_1, v_2, \dots, v_n\}$ of the orthonormal bases obtained in (2). Then E is also an orthonormal basis for $M_{n \times 1}(\mathbb{R})$.
4. $D = P^\top AP$ where $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ and $P = [v_1 \ v_2 \ \cdots \ v_n]$.

Example 14.3. Let $A = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$. Find an orthogonal matrix P and diagonal matrix D such that $D = P^\top AP$.

Solution.

Find the characteristic polynomial $p_A(x)$.

$$p_A(x) = \begin{vmatrix} -x & 2 & 2 \\ 2 & -x & 2 \\ 2 & 2 & -x \end{vmatrix} = -(x+2)^2(x-4)$$

Therefore, the eigenvalues of A : $-2, 4$

- $E(A, -2)$

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ 2 & 2 & 2 & 0 \\ 2 & 2 & 2 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a + b + c = 0 \end{cases}$$

Let $b = s$ and $c = t$, then, $a = -s - t$,

$$E(A, -2) = \left\{ \begin{bmatrix} -s-t \\ s \\ t \end{bmatrix} \mid s, t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Because they are not scalar multiples of each other, it is a basis for $E(A, -2)$.

1. Put $w_1 = v_1$,

$$w_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

And since $\|w_1\| = \sqrt{2}$,

$$w'_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

2. Put $w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$,

$$w_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

And since $\|w_2\| = \sqrt{\frac{3}{2}}$,

$$w'_2 = \sqrt{\frac{2}{3}} \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

- $E(A, 4)$

$$\left[\begin{array}{ccc|c} -4 & 2 & 2 & 0 \\ 2 & -4 & 2 & 0 \\ 2 & 2 & -4 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \longleftrightarrow \begin{cases} a - c = 0 \\ b - c = 0 \end{cases}$$

Let $c = t$, then, $a = t, b = t$,

$$E(A, 4) = \left\{ \begin{bmatrix} t \\ t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

And since it contains a nonzero element it is a basis for $E(A, 4)$. However, we should still normalize this. Since $\|w_1\| = \sqrt{3}$, then,

$$w'_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Therefore,

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \quad P = \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$